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Paleomagnetic and geochemical studies of the Mesoproterozoic Satakunta dyke swarms, Finland, with implications for a Northern Europe – North America (NENA) connection within Nuna supercontinent

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ABSTRACT

The existence of the Paleoproterozoic supercontinent Nuna (Columbia, Hudsonland) has been proposed by several authors. Many recent reconstructions of this supercontinent are based on the assumption that Baltica and Laurentia form part of its core in a Northern Europe – North America (NENA) configuration. The E–W, N–S and NE–SW trending Early Mesoproterozoic diabase dyke swarms in Satakunta, SW Finland, provide new paleomagnetic data to test the NENA configuration. A high-inclination, secondary remanence component carried by maghemite was isolated from some sites, and is interpreted to record hydrothermal alteration during the initial Mesozoic breakup of Pangea. A dual-polarity, high-stability remanence component was found in about half of the studied sites, and confirmed to be primary by several positive baked-contact tests. A suite of rock-magnetic analyses demonstrates that pseudo-single domain magnetite is the remanence carrier. The combined mean direction for N–S (1565 Ma; Lehtonen et al., 2003) and NE–SW trending dykes, showing both polarities, is $D = 11.5^\circ$, $I = 3.3^\circ$ ($\alpha_{95} = 8.8^\circ$), yielding a key paleomagnetic pole (SK1) for Baltica at 29.3° N, 188.1° E ($A_{95} = 6.6^\circ$). It fulfills the first six of seven Van der Voo's (1990) reliability criteria for paleomagnetic poles. The new Satakunta SK1 pole, when compared to the nearly coeval Western Channel Diabase pole (1590 ± 4 Ma) from Laurentia, allows the NENA fit at 1.57–1.59 Ga. Based on geological evidence E–W dykes have been proposed to belong a swarm that is ca. 100 million years older than the N–S dykes. E–W trending dykes show a dual polarity direction of $D = 356.9^\circ$, $I = 8.3^\circ$ ($\alpha_{95} = 15.9^\circ$), yielding a paleomagnetic pole (SK2) for Baltica at 32.6° N, 205.5° E ($A_{95} = 14.3^\circ$). Based on coeval paleomagnetic data and correlations of geochronology and basement geology of Baltica and Laurentia, the NENA fit is validated at 1.77–1.75 Ga, 1.59–1.57 Ga, 1.46 Ga, and 1.27 Ga, and by comparing single virtual geomagnetic poles at 1.63 Ga. However, the mean 1.63 Ga data from Laurentia (Melville Bugt dykes, Greenland; Halls et al., 2011) and Baltica (Sipoo dykes, Finland) are offset by about 30°. Validation of NENA for both older and younger times suggests to us that a variety of factors, such as less than highest-quality or not coeval paleomagnetic data, possible unrecognized tilting of the continental blocks, or a non-symmetric geomagnetic field, other than cratonic reconstruction, can explain the minor 1.63 Ga mean pole discordance.

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1. Introduction

The concept of supercontinent cyclecity is established mainly based on the global peaks in distribution of zircon U–Pb and whole rock Nd-model ages (Condie, 1998, 2004). Reddy and Evans (2009) recently summarized the secular evolution of Earth from the Neoarchean to the Mesoproterozoic Eras, revealing intriguing temporal relationships of the supercontinent cycles with evolution of Earth's core, mantle, crust, oceans, atmosphere, and life. For example, the temporal distribution of rapakivi magmatism worldwide at 2.8–2.6 Ga, 1.8–1.0 Ga, and 1.0–0.5 Ga could be coeval with supercontinent cycles (e.g. Rämö and Haapala, 1995; Condie, 1998; Åhäll et al., 2000; Haapala et al., 2005; Larin, 2009). In southeastern Fennoscandia, the rapakivi granite batholiths represents the silicic member of a bimodal magmatic complex, in which tholeiitic diabase dykes and minor gabbroic-anorthositic bodies

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represent the mafic member (Lindberg and Eklund, 1992; Haapala and Rämö, 1992; Eklund et al., 1994). Paleomagnetism of these Late Mesoproterozoic formations that are commonly called Subjotnian (ca. 1.65–1.54 Ga) in Fennoscandia have been widely-studied (e.g. Neuvonen and Grundström, 1969; Neuvonen, 1970, 1978, 1986; Bylund, 1985; Pesonen, 1987; Pesonen et al., 1987, 1991; Bylund and Elming, 1992; Mertanen and Pesonen, 1995; Mertanen et al., 2008). However, a closer look at these data shows that many of them are scattered with large and overlapping 95% confidence circles (A_{95}) about the calculated mean paleomagnetic poles, and only a few of these poles come from rock formations that have been radiometrically dated and whose magnetization has been proven to be primary. Our aim of this study is to fill in the gaps of the paleomagnetic record with a new, well-defined Subjotnian paleomagnetic pole for Baltica.

A basic assumption for paleomagnetic research is that when averaged over a sufficient geological time interval ($\sim 10^4$ to 10^5 years), the Earth's magnetic field may be approximated by geocentric axial dipole (GAD, e.g. Hospers, 1954; Irving, 1964, 2005). However, its existence during Precambrian must be tested (e.g. Evans, 2006). Support for dynamo activity since the Late Archean–Early Proterozoic began when Smirnov and Tarduno (2004) pointed out that the magnitude of secular variation at ca. 2.5 Ga (recorded by dykes from Superior and Karelia) was already similar to the present one. More recent work by Biggin et al. (2008) and Smirnov et al. (2011) showed that the dynamo was already working at 2.5 Ga, and although geomagnetic secular variation during the Late Archean and Early Proterozoic was different from that of the past 200 million years, the Mid-Proterozoic interval had already become similar to the modern. Moreover, Evans (2006) showed that the GAD model fits the distribution of evaporite basins on Earth throughout most of the last two billion years, indicating that GAD should be valid for Subjotnian times.

The existence of the Paleoproterozoic supercontinent Nuna (Columbia, Hudsonland) has been proposed by several previous studies (e.g., Williams et al., 1991; Hoffman, 1996; Meert, 2002; Rogers and Santosh, 2002; Zhao et al., 2002; Pesonen et al., 2003; Condie, 2004). Baltica (East European craton) and east Laurentia (North America and Greenland) are thought to represent two of the most important building blocks of this supercontinent. The temporal and compositional overlap of the anorogenic and orogenic magmatism between west Baltica and east Laurentia, as well as the paleomagnetic data available, suggest that these continents coexisted for more than 600 Ma (from ca. 1.2 to 1.8 Ga), constituting so called 'NENA' (Northern Europe – North America; Gower et al., 1990). A rather limited paleomagnetic dataset in the last century supported a NENA-like Mesoproterozoic reconstruction (Patchett et al., 1978; Piper, 1980) that was either static (Buchan et al., 1990) or internally shearing (Pesonen et al., 2003). More recent data have appeared to support, within the analytical uncertainties, a single NENA juxtaposition between Baltica and Laurentia between ca. 1750 and ca. 1270 Ma (Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008; Lubnina et al., 2010; Pisarevsky and Bylund, 2010; Evans and Mitchell, 2011; Pesonen et al., 2012), and perhaps lasting as long as ca. 1120 Ma (Salminen et al., 2009; Raiskila et al., 2011) and forming the core of Nuna. Moreover, Nuna might include other continents, such as Siberia (e.g. Wingate et al., 2009; Evans and Mitchell, 2011), Amazonia (Bispo-Santos et al., 2008, 2012; Johansson, 2009), Australia (e.g. Hoffman, 1991; Karlstrom et al., 2001; Betts et al., 2008; Payne et al., 2009), and North China (Halls et al., 2000; Wu et al., 2005; Kusky et al., 2007; Zhang et al., 2012). Many of these reconstructions are based on the assumption that Baltica and Laurentia form the core of Nuna in a configuration similar to NENA. However, recent paleomagnetic data from ca. Melville Bugt dykes (1.63 Ga) in Greenland does not support that fit (Halls et al., 2011). To address these concerns, we present here new key and other well defined paleomagnetic pole from the Satakunta mafic dyke swarms (one dyke U–Pb dated at 1565 Ma; Lehtonen et al., 2003) from the West coast of southern Finland and suggest that NENA was also valid at 1.63–1.57 Ga. Based on the trending at the field, geochemical and paleomagnetic data we divided studied Satakunta Subjotnian dykes in two swarms: (1) E–W and (2) combined N–S and NE–SW.

2. Geological setting

Most continental crusts in the southern Finland were formed during the Svecofennian (a.k.a. Svecokarelian) orogeny at ca. 1.9–1.8 Ga (Huhma, 1986), related to an oblique collision of a growing Svecofennian island arc system against the Archean craton and to the subsequent continental deformation, metamorphism and crust-forming magmatism (e.g., Korja et al., 2006; Lahtinen et al., 2008). The post-collisional uplift at 1.8 Ga ended the Svecofennian orogeny. About 200 Ma later, the region entered into phases of a long period of crustal extension, development of rift basins, and anorogenic magmatism, including intrusion of several mafic dykes and sills (Eklund and Shebanov, 2005; Andersson et al., 2006). These magmatic units have been traditionally divided into three main groups based on their age relationship to the rift-filling ("Jotnian") sandstones: Subjotnian (ca. 1.65–1.54 Ga), Jotnian, and Postjotnian (ca. 1.25 Ga). Subjotnian units are exposed in the southern part of Finland (Fig. 1) such that the younger units seem to be exposed in the SW part of Finland (e.g. Åland, Vehmaa, and Laitila) and the older in the SE part of Finland (e.g. Wiborg, Suomenniemi, and Häme; Fig. 1). However a recent detailed mineralogical study of ovoids from the Vehmaa rapakivi (SW Finland; Fig. 1) batholith by Shebanov et al. (2000) showed that core zones of these ovoids give a U–Pb (zircon) age of 1630 Ma, and the matrix gives a U–Pb (zircon) age of 1573 Ma. These data indicate that in SW Finland there is a deep Subjotnian assemblage that gives the same age as the E–W dyke swarm and the SE Finnish rapakivi batholiths, proposed as a continuous Subjotnian magmatic province (Shebanov et al., 2000). Furthermore, both chemical and mineralogical compositions provide evidence for magma mixing between the rapakivi and mafic magmas, indicating that the emplacement of Subjotnian rapakivi batholiths were coeval with the intrusion of the dykes (Lindberg and Eklund, 1992; Eklund et al., 1994).

In Satakunta, the Svecofennian crust consists of Paleoproterozoic granitoids, paragneisses and felsic and intermediate metavolcanic rocks (Fig. 2). This assemblage is cut by several Subjotnian rapakivi granite intrusions and mafic dyke swarms trending E–W, N–S and NE–SW belonging in different age groups. We did not observe and there are no reports of Postjotnian diabase dykes exposed near our sampling sites (e.g. Pihlaja, 1987; Lehtonen et al., 2003; Suominen et al., 2006). Based on petrography, mode of occurrence, and field relations, it has been suggested that the Subjotnian dykes in Satakunta are continuations of both the ESE–WNW trending Häme dyke swarm (1646 ± 6 Ma: U–Pb (zr), Laitakari, 1987 and 1667 ± 9 Ma: Vaasjoki and Sakko, 1989) and the SSW–NNE trending Föglö dyke swarm (1577 ± 12 Ma: U–Pb (bd + zr) and 1540 ± 12 Ma, Suominen, 1991) (Fig. 1). The widths of the Subjotnian swarms are tens of kilometers, and the lengths are likely ca. 300 km. The dykes dip vertically or subvertically. Dyke widths vary from a few centimeters to over one hundred meters. One of the presently studied N–S trending dykes has a baddeleyite U–Pb age of ca. 1565 Ma (Lehtonen et al., 2003), suggesting that it belongs to the same system as the Föglö dykes. Lehtonen et al. (2003) obtained discordant ($^{207}Pb/^{206}Pb$) age results with three age series of ~1476, 1513, and 1565 Ma. Since the two first results were so close to each other, basically forming one point, no error estimation was possible. The

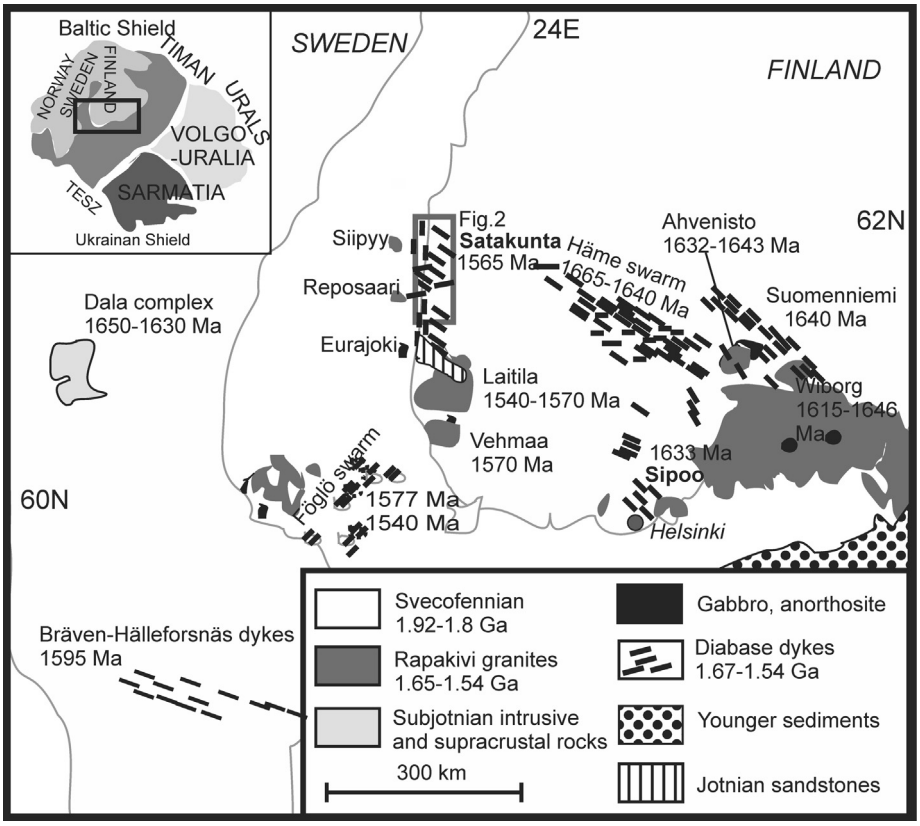


Fig. 1. Simplified geological map of Southern Fennoscandian shield.
Modified after Rämö (1991).

zircon upper intercept age is ca. 1.59 Ga, supporting the Subjotnian age for the dyke. Lehtonen et al. (2003) interpreted that the $^{207}Pb/^{206}Pb$ age of 1565 Ma best represents the time of crystallization.

Pihlaja (1987) and Lehtonen et al. (2003) studied petrography of these dykes. The major mineral phases include plagioclase, hypersthene, augite, and olivine. Accessory minerals are apatite, ilmenite, magnetite, and sometimes sulfides. Some of the Subjotnian dykes in Finland are highly altered with hydrothermal mineral replacement (e.g. epidote, calcite, and sericite) (Ehlers and Ehlers, 1977; Pihlaja, 1987; Mänttäri et al., 2005; Paulamäki et al., 2006; Mertanen, 2008).

3. Samples and methods

3.1. Sampling

Altogether, forty-one diabase dyke sites were sampled at Satakunta, Finland, spanning ca. 100 km from the northernmost sampling site to the southernmost sampling site (Fig. 2). The sampling area is located north of the Satakunta sandstone (see Klein et al., in this volume) and Postjotnian mafic intrusions. From thirty-one sites we sampled baked host rocks, and from twenty-two of them the unbaked host rock was sampled for baked contact tests to study the origin of magnetic remanence (Everitt and Clegg, 1962). In thirty-one of the sites, standard oriented cores were collected with a portable field drill, in seven of the sites both oriented core and block samples were taken, and in three sites only oriented block samples were taken. Cored samples were oriented using both sun and magnetic compasses, whereas block samples were oriented using only a magnetic compass.

3.2. Magnetic measurements

Paleomagnetic measurements were conducted in the magnetically shielded room of the paleomagnetic laboratory of the Department of Geology and Geophysics at the Yale University, USA. After the measurement of natural remanent magnetization (NRM), samples were placed into liquid nitrogen in a null field to demagnetize viscous remanent magnetization (Borradaile et al., 2004). Thereafter, the majority of samples were thermally demagnetized to separate the characteristic remanent magnetization (ChRM) component. A few sister specimens were stepwise demagnetized using the alternating field (AF) method. The thermal method turned out to be more effective and provided more stable results than the AF method. Remanent magnetization was measured using an automated sample-changing system attached to a 2G cryogenic magnetometer (Kirschvink et al., 2008). A nitrogen-atmosphere ASC Scientific model TD-48SC furnace was used for thermal demagnetization. The demagnetization results were analyzed using orthogonal plots, stereographic projections, and demagnetization decay curves (Zijderveld, 1967) and calculated using principal component analysis (Kirschvink, 1980; Jones, 2002). Components with a minimum of four demagnetization steps were used to calculate vector lines using least-square fits, and an upper limit for maximum angular deviation (MAD) less than 6° was used as a quality filter. In a few samples directions with MAD up to 10° were accepted (from sites KA, OJ, PL, RK, SA, VR) for components interpreted to be overprints. Mean remanence directions were calculated using Fisher statistics (Fisher, 1953). Data was not streaked toward any overprinted directions except slightly in sites AM and AT, but results obtained using Fisher statistics are similar as those using Bingham statistics (Onstott, 1980).

Magnetic mineralogy was investigated by several methods using selected powdered whole-rock samples. Susceptibility

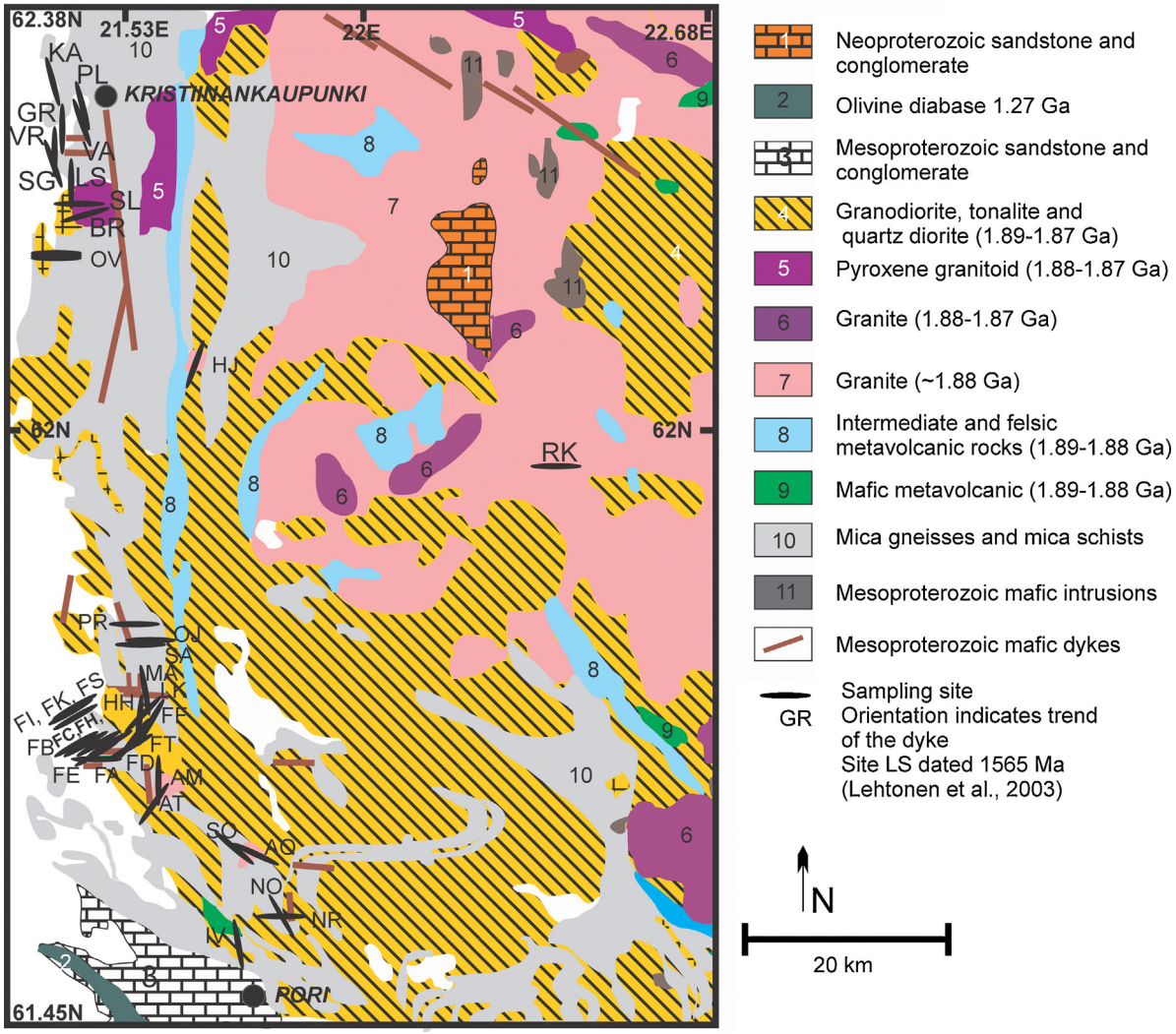


Fig. 2. Geological map of the study area with the location of sampling sites.
Modified after Korsman et al. (1997).

measurements at both low and high temperature were done at Yale University using AGICO's KLY4-CS Kappabridge instruments to characterize magnetic carriers in the samples. The low-temperature experiment, in which samples were heated from −194 °C to room temperature (RT, 25 °C), was carried out first. Then, the same samples were heated from RT to 700 °C and cooled back to RT in argon gas which helped to avoid oxidation of minerals during the high-temperature experiments. AGICO's Kappabridge was used to measure the anisotropy of magnetic susceptibility (AMS). The technique of Jelinek (1981) was used for statistical analyses of AMS data. Room-temperature hysteresis properties were measured to determine domain states of magnetic carriers at the Institute for Rock Magnetism (IRM), University of Minnesota, USA, using a vibrating sample magnetometer (from Princeton Measurements Corporation). Low-temperature remanence measurements to explore magnetic carriers were conducted at IRM using a magnetic properties measurement system (MPMS) superconducting quantum interference device (SQUID) magnetometer (from Quantum Design, Inc.).

3.3. Geochemical sample processing and measurements

Trace element, rare earth element (REE), and Pb-isotope compositions of bulk samples were analyzed at Yale University. Small rock chips from the fresh parts of samples in the middle of the unheated paleomagnetic core were ground using hand operated iron mill. All sample processing was performed in a clean room of the Yale University. About 0.1–0.15 g of sample powders were weighed and dissolved at 100 °C for over 12 h in a high-pressure digestion system using 4:1 HF + $HClO_4$ + trace amount of HNO_3 in a Teflon beaker under the pressure about 2 bar. After total dissolution the sample solution was dried by slowly raising the temperature from 100 to 210 °C, in order to evaporate residual HF, SiF_4, and $HClO_4$. After the samples are completely dried, 6.2 N HCl was added and samples were dried again at 210 °C. This procedure was repeated three times, during the last time of which, a drop of H_2O_2 was added to adjust oxygen fugacity. After samples are completely dried, 4.0 ml of 6.2 N HCl was used to dissolve the sample.

A small aliquot of the samples were dried down again, nitrified by 3 N HNO_3, and then dissolved in 5% HNO_3 with 2 ppb Indium (as an internal standard) for measuring trace metal and REE element concentrations using ICP-MS (Element-XR) at Yale University. The reference material BHVO-2 (basalt from Pahoehoe lava from Halemaumau, Hawaii, USA) issued by United States Geological Survey (USGS) was used as a standard. The typical precision for trace metal and REE elements analyses is better than ±3%.

An aliquot of samples (about 0.08–0.1 g) were processed for Pb-isotope analyses. All samples were first passed through an

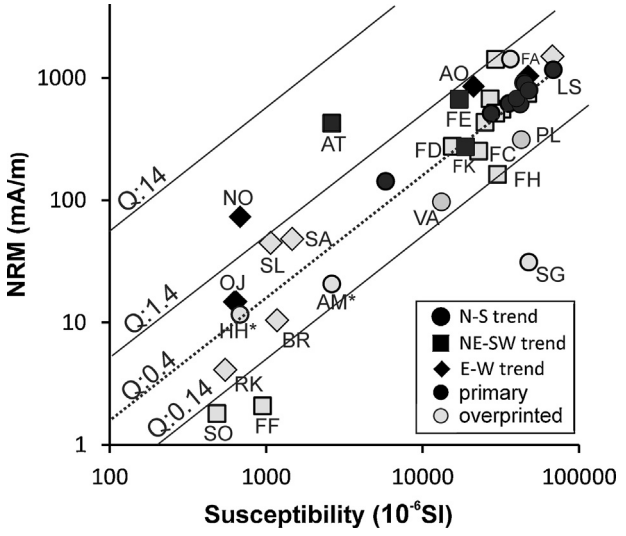


Fig. 3. Remanence-susceptibility (logarithmic scale) diagrams for Satakunta (Finland) dyke samples. Koenisgsberger's Q ratios (calculated with the field value of 50 μT) are shown as straight lines. Dykes showing shallow primary component P are marked with black and dykes showing overprint secondary component S are marked with gray. *Dykes from site have been divided to two groups and this is the group with lower NRM values (see Table 1).

anion exchange column to separate Fe from other elements (AG1-X8 anion resin, Biorad column, 0.5 cm OD and 3.0 cm long). Pb isotopes were separated from the portion without Fe using the HBr–HNO_3 procedure similar to Abouchami et al. (1999). The typical blank for this protocol is <5 pictogram. Pb isotope compositions were measured using multi-collector ICP-MS (Neptune) at Yale University. The SRM997 Tl was used as an internal standard for mass-bias correction ($^{203}Tl/^{205}Tl = 0.418923$), and typical external reproducibility (including full chemistry) are ~12 ppm (2σ) for $^{207}Pb/^{206}Pb$, ~16 ppm (2σ) for $^{208}Pb/^{206}Pb$, ~0.002 (2σ) for $^{206}Pb/^{204}Pb$, ~0.0016 (2σ) for $^{207}Pb/^{204}Pb$, and ~0.004 (2σ) for $^{208}Pb/^{204}Pb$, comparable with other studies (e.g., Hart and Blusztajn, 2006).

4. Results

4.1. Petrophysics and paleomagnetism

Petrophysical properties are presented in Table 1 and Fig. 3. The intensity of the remanence of dyke samples ranges from 2 to 1500 mA/m. Susceptibility of the samples ranges from 550 × 10^{-6} SI to 68,600 × 10^{-6} SI. The wide range of remanence intensity and susceptibility values is probably caused by variation in the grain sizes and/or degree of weathering. Koenigsberger's Q value is the ratio of the remanent magnetization to the induced magnetization in the Earth's magnetic field. Q values of Satakunta dykes range from 0.1 to 3 (Fig. 3 Q values lower than 0.4 are commonly associated with overprinted or unstable magnetization).

Stable paleomagnetic directions were obtained from thirty-two sites (Table 2). We identified three components from the Subjotnian dyke samples: (1) a shallow north component and its reversed direction with high unblocking-temperature hereafter called 'primary component P' (Figs. 4 and 5); (2) a steep downward ENE component hereafter called secondary component S (Fig. 6); and (3) a steep downward and northerly component of viscous origin that shows Present day Earth's magnetic Field (PEF) direction ($D = 7°$, $I = 74°$). Component P was obtained from seventeen sites, among which six samples also show component S (i.e. AM, AO, FA, FE, IV, MA). In these cases component S was observed only in samples near the centers of the dykes or further away from the margins

Table 1
Site mean petrophysical properties of the Satakunta mafic dykes.

Site	*n*	NRM (mA/m)	Susceptibility (10^{-6} SI)	Q	Site	*n*	NRM (mA/m)	Susceptibility (10^{-6} SI)	Q	Site	*n*	NRM (mA/m)	Susceptibility (10^{-6} SI)	Q
N–S trending dykes					NE–SW trending dykes					E–W trending dykes				
AM[a]	8	21 ± 16	2635 ± 2150	0.2 ± 0.04	AT	4	110 ± 1018	2481 ± 7	0.7 ± 0.28	AO	6	854 ± 243	21220 ± 1024	0.9 ± 0.32
AM	6	910 ± 398	44137 ± 16522	0.5 ± 0.22	FB	5	555 ± 250	32228 ± 5158	0.3 ± 0.07	BR	5	10 ± 3	1179 ± 60	0.2 ± 0.07
GR	6	660 ± 314	40363 ± 9248	0.4 ± 0.18	FC	3	253 ± 167	22857 ± 4684	0.3 ± 0.22	FA	8	1039 ± 301	47470 ± 8503	0.5 ± 0.16
HH[a]	2	12 ± 3	681	0.4 ± 0.1	FD	6	277 ± 188	15530 ± 10000	0.3 ± 0.08	NE, NI, NM	11	15 ± 15	645 ± 92	0.6 ± 0.54
HH	5	143 ± 101	5817 ± 2184	0.9 ± 0.42	FE	7	671 ± 507	17168 ± 12693	±0.65	NO	8	73 ± 70	684 ± 65	3.1 ± 2.51
HJ	15	1425 ± 644	36408 ± 7035	0.9 ± 0.42	FF	3	2 ± 265	949 ± 762	±2.7	NR	8	1501 ± 532	67684 ± 13568	0.6 ± 0.28
IV	10	514 ± 104	27371 ± 5177	0.4 ± 0.09	FH	3	163 ± 72	30190 ± 5670	0.2 ± 0.05	OJ	6	15 ± 5	627 ± 195	0.7 ± 0.34
KA	6	747 ± 203	46998 ± 4894	0.4 ± 0.08	FI	6	677 ± 258	26970 ± 3427	0.6 ± 0.17	RK	7	4 ± 2	550 ± 37	0.2 ± 0.08
LS	5	1170 ± 272	68596 ± 4171	0.4 ± 0.09	FK	13	274 ± 201	18805 ± 22019	0.7 ± 0.39	SA	5	48 ± 22	1470 ± 140	0.7 ± 0.37
MA	9	797 ± 275	47997 ± 1223	0.4 ± 0.12	FS	7	434 ± 176	25065 ± 3985	0.4 ± 0.08	SL	5	45 ± 11	1073 ± 60	1.0 ± 0.25
OV	9	618 ± 250	35690 ± 14316	0.4 ± 0.12	FT	13	517 ± 305	29351 ± 12627	0.5 ± 0.22					
PL	7	313 ± 186	42713 ± 9121	0.2 ± 0.09	LK	6	1422 ± 426	29310 ± 10130	1.7 ± 1.07					
SG	5	31 ± 9	47698 ± 5671	0.01 ± 0.01	SO	4	2 ± 1	485 ± 33	0.1 ± 0.04					
VA	6	97 ± 55	13202 ± 2526	0.1 ± 0.08										
VR	5	614 ± 157	42078 ± 6658	0.3 ± 0.04										

n is the number of samples; Q is the Koenigsberger's ratio calculated for a field of 50 μT; the standard deviations are for each site.
[a] These samples showed one to two order of magnitude lower NRM values than other samples from the same site.

Table 2
Paleomagnetic results from Satakunta dykes and host rocks.

Site	Rock type	Lat(° N)/logn(° E)	Str(°)/dip(°)	Width (m)	D (°)	I (°)	α_{95} (°)	k	Plat (° N)	Plong (° E)	A_{95} (°)	K	n	Pol
Primary component P														
N–S														
AM	Dyke	61.662/21.620	360/90	>20	2.6	−4.2	20.3	8.4	26.1	198.7	11.3	24.9	8	N
HJ	Dyke	62.059/21.670	20/90	10	15.9	17.0	9.0	19.2	35.5	182.0	8.7	20.4	15	N
IV	Dyke	61.542/21.752	350/90	20	27.2	18.7	18.7	17.6	34.4	168.2	18.3	18.4	5	N
LS	Dyke	62.206/21.429	360/?	>60	33.9	36.1	10.9	49.8	42.4	156.4	8.7	77.4	5	N
MA	Dyke	61.726/21.609	350/90	15	22.1	15.0	15.2	14.3	33.9	174.4	11.8	22.8	8	N
AM	Baked granite	61.662/21.620			21.4	−6.0			23.3	178.2			2	N
HJ	Baked granite	62.059/21.680			22.8	6.5	24.0	15.6	28.8	175.4	21.9	18.6	4	N
IV	Baked gneiss	61.542/21.752			28.0	25.5	42.2	9.6	38.1	167.1	43.6	9.1	3	N
GR[a]	Dyke	62.240/21.380	360/60	1.5	171.2	−12.1	22.5	12.5	−33.6	32.0	22.1	12.9	5	R
OV1[a]	Dyke	62.154/21.395	10/60	0.15	186.9	36.2	25.9	23.7	−6.7	15.6	21.9	32.6	3	R
OV2[a]	Dyke	62.154/21.395	10/60	0.05	188.7	13.1	12.3	101.1	−20.8	12.1	11.3	120.7	3	R
OV3[a]	Dyke	62.154/21.395	10/60	0.015	189.0	13.8	11.4	117.2	−20.4	11.9	6.9	324.1	3	R
GR[a]	Dyke	62.240/21.380	360/60	1.5	171.2	−12.1	22.5	12.5	−33.6	32.0	22.1	12.9	5	R
GR[a]	Baked gneiss	62.240/21.380			183.2	3.6			25.9	197.8			1	R
OV[a]	Baked gneiss	62.154/21.395			174.8	5.1			−25.2	27.1			1	R
Mean N–S					13.1	3.7	11.4	13.0	29.2	186.2	8.2	24.6	15*/71	N, R
NE–SW														
AT	Dyke	61.644/21.635	35/90	0.1	348.7	12.8	30.0	10.3	34.7	216.1	21.5	19.3	4	N
FE	Dyke	61.679/21.545	60/72	0.3	28.3	3.1	29.3	10.8	26.4	169.6	22.6	17.5	4	N
AT	Baked granite	61.644/21.635			8.3	−3.5			26.3	192.4			1	N
Additional sites from Mertanen, 2008														
DS_p	Dyke	61.243/21.479	60/70	1.5	5.5	−0.1	6.4	90.6	28.4	195.2	5.6	116.2	1/7*/17	N
DB_p	Dyke	61.224/21.508	66/90	0.06	4.6	−0.7	13.9	19.8	28.6	196.2	10.7	32.9	1/7*/7	N
Mean NE–SW					6.8	3.9	20.0	22.1	30.3	193.7	19.0	24.2	5*/33	N
SK1: mean N–S and NE–SW				11.5	3.3	8.8	15.5	29.3	188.1	6.6	26.6	20*/104	N, R	
E–W														
AO	Dyke	61.607/21.783	295/60	0.25	12.8	5.3	7.1	73.3	30.3	186.9	6.3	94.2	7	N
NO	Dyke	61.560/21.820	330/65	0.6	349.7	8.4	35.5	7.7	32.2	214.1	35.4	7.7	4	N
OJ	Dyke	61.744/21.616	270/90	0.1	351.5	21.8	4.9	58.1	40.6	212.7	5.0	55.2	16	N
AO	Baked granite	61.607/21.783			10.5	−1.0	23.2	11.8	27.4	190.0	21.6	13.5	5	N
FA[a]	Dyke	61.679/21.545	270/90	0.35	159.4	−6.2	33.7	14.5	−29.6	44.9	29.4	18.6	3	R
SK2: mean E–W				356.9	8.3	15.9	24.2	32.6	205.5	14.3	29.8	5*/35	N, R	
Unbaked SVF direction														
AM	Granite	61.662/21.620			334.6	60.2			65.5	250.3			2	N
FA	Migmatite	61.679/21.545			354.6	49.7			58.6	210.5			1	N
HJ	Granite	62.059/21.690			356.1	49.3			58.0	208.0			1	N
SL	Granitoid	62.197/21.441			318.6	34.8			38.4	254.3			1	N
Secondary component S														
AM	Dyke	61.662/21.620	360/90	>20	033.5	56.0	12.4	13.2	57.4	146.3	14.5	9.9	12	N
AO	Dyke	61.607/21.783	295/60	0.25	27.3	47.1	41.9	3.0	52.1	160.7	42.8	2.9	7	N
FA	Dyke	61.679/21.545	270/90	0.35	40.3	59.3	29.2	3.4	57.5	134.3	38.1	2.4	11	N
FB[b]	Dyke	61.683/21.550	240/?	1	353.8	69.3	17.1	16.4	80.6	225.1	26.9	7.2	6	N
FC	Dyke	61.680/21.550	60/90	7	92.4	59.8	41.2	5.9	34.9	095.7	60.5	3.3	4	N
FD	Dyke	61.680/21.546	60/50	0.3	84.6	57.0	38.3	6.7	66.3	064.7	54.0	3.9	4	N
FE	Dyke	61.679/21.545	60/72	0.3	022.4	76.1	19.4	7.2	79.6	090.8	30.9	3.4	10	N
FH	Dyke	61.681/21.546	60/80	0.3	57.4	80.2	19.3	41.9	29.6	060.0	35.6	13.0	3	N
FI	Dyke	61.687/21.594	60/90	0.2	124.1	66.7	12.2	31.0	29.1	104.8	18.4	14.3	6	N
FK	Dyke	61.680/21.542	58/90	0.8	80.8	47.3	35.5	13.1	34.0	087.6	42.7	9.4	3	N
FS	Dyke	61.680/21.542	50/90	0.35	102.5	58.7	16.2	14.9	28.7	080.8	22.3	8.3	7	N
IV	Dyke	61.542/21.752	350/90	20	24.0	62.1	18.5	8.7	66.9	152.7	26.2	4.8	9	N
MA	Dyke	61.726/21.609	350/90	15	43	52.8	29.4	10.8	50.7	137.6	42.6	5.6	4	N

Table 2 (*Continued*)

Site	Rock type	Lat(° N)/logn(° E)	Str(°)/dip(°)	Width (m)	D (°)	I (°)	α_{95} (°)	k	Plat (° N)	Plong (° E)	A_{95} (°)	K	n	Pol
NR	Dyke	61.560/21.820	275/85	0.2	85.1	46.1	35.3	2.3	26.2	102.0	38.1	2.1	13	N
RK	Dyke	61.975/22.333	270/90	4	53.0	62.8	11.5	45.5	54.9	117.6	16.7	22.0	5	N
SA	Dyke	61.744/21.616	270/75	?	56.3	83.4	20.3	9.8	66.5	49.7	34.6	4.0	7	N
VA[c]	Dyke	62.269/21.434	340/90	15	33.6	29.3	15.8	24.5	37.8	159.1	12.2	40.4	5	N
Mean					62.6	64.3	8.9	19.4	53.3	103.9	13.2	9.4	15*/110	N

lat/long – sampling latitude and longitude, Str/dip – strike and dip of the dyke, width – width of dyke, D – declination, I – inclination, α – the radius of the 95% confidence cone in Fisher (1953) statistics, Plat/Plong – paleolatitude/paleolongitude of the pole, A_{95} – radius of the 95% confidence cone of the VGP, K – Fisher precision parameter, (N)/n – number of analyzed (sites)/samples, * – the number used to calculate mean value, pol – polarity of the isolated direction: N(R) – normal (reversed) polarity.

[a] Reversed directions are inverted for grand mean. SVF – Svecofennian.

[b] Not in mean.

[c] These are originally of Svecofennian aged dykes.

(larger grain size) of host rocks and dykes. One site LS (age 1565 Ma; Lehtonen et al., 2003) gave a remanence direction between P and the PEF/S-clusters (Fig. 7). Stable component S and/or PEF were obtained from seventeen sites. Three sites (SL, RK, and VR) show WNW downward directions close to the well-known Svecofennian paleomagnetic direction. Due to the higher degree of metamorphism at site SL, it was noted in the field that the dyke may be of Svecofennian age. Since in this study we are concentrating on Mesoproterozoic dykes, these results are not further discussed. It is worth noting that one sample from dyke VR shows component P as an overprint on the Svecofennian direction, which we speculate as a possible thermochemical overprint by a nearby Mesoproterozoic dyke that is not exposed.

4.1.1. Primary component P

In general, similar characteristic remanence directions with unblocking temperatures from 500 °C to 585 °C for component P were isolated from all dykes regardless of the dyke trend. Both N–S and E–W trending dykes show two polarities whereas NE–SW trending dykes show only normal-polarity directions. We employed the standard convention of assigning "normal" polarity to north-directed ChRM vectors from Proterozoic rocks in Baltica; relating this arbitrary definition to the absolute sense of Mesoproterozoic geomagnetic polarity is uncertain due to the lack of continuity in the Baltica APWP through Neoproterozoic ages. The trajectories in Zijderveld diagrams pass through the origin, indicating that there is no higher-temperature component present in the samples.

Full baked contact tests were obtained from two sites (AM and HJ) of N–S trending dykes (Figs. 8 and 9, respectively). At these sites, dykes and baked host rocks gave component P, and unbaked Svecofennian-aged host rocks showed the expected Svecofennian direction. Unblocking temperatures for component P in baked host rocks are 450–585 °C. In addition to these fully constrained baked contact tests, component P was obtained at several sites (AO, AT, GR, IV, and OV) from baked host rocks (e.g. Fig. 4), but unbaked host rocks at these sites were unstable during demagnetization. Reversed polarity components were obtained for both dykes and baked host rocks from sites OV and GR (for example Fig. 4c and d). Table 2 presents a summary of paleomagnetic results, and site mean directions for component P are shown in Fig. 10. Because field observations indicate that E–W trending dykes are older than N–S and NE–SW trending dykes, we have combined mean values for component P in respect to the trend of dykes. The combined mean direction for N–S and NE–SW trending dykes, plus two additional NE–SW sites from Mertanen (2008), has parameters D = 11.5°, I = 3.3° (20 sites, α_{95} = 8.8°) and yields a dual-polarity paleomagnetic pole located at Plat = 29.3° N, Plong = 188.1° E (A_{95} = 6.6°). The mean value for E–W trending dykes is D = 356.9°, I = 8.3° (5 sites, α_{95} = 15.9°) and yields a dual-polarity paleomagnetic pole located at Plat = 32.6° N, Plong = 205.5° E (A_{95} = 14.3°).

At the dated site LS (1565 Ma, Lehtonen et al., 2003) the interior of the dyke (with a width > 60 m) was sampled since the contacts were not exposed. Due to large mineral grains, more than half of the NRM intensity was lost during the liquid nitrogen treatment. At the beginning of demagnetization a strong PEF direction was observed, which persisted until 560 °C. Then, an intermediate steep direction was observed before samples were fully demagnetized at 583 °C (Fig. 7), which is somewhat steeper than the P component separated from the other dykes, but within a reasonable expectation for the secular variation of the geomagnetic field. Thus, the site is included in the P mean calculations (Fig. 10).

4.1.2. Secondary component S

A steep-down, secondary component S with ENE declination was isolated from half of the sites, in temperature ranges from

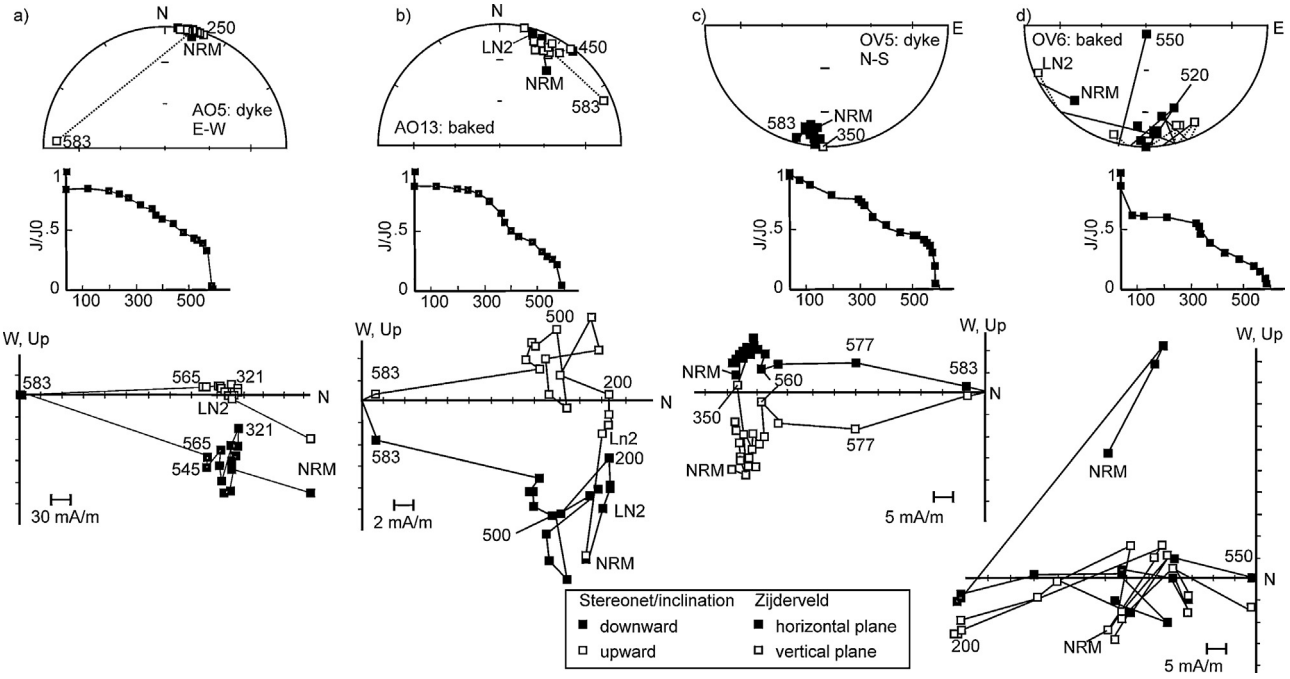


Fig. 4. Example of thermal demagnetization of E–W trending dyke showing primary component P. Figures show stereographic projection, normalized intensity curve (J/J_0) and orthogonal projections of remanent magnetization directions. (a) Normal polarity direction obtained from dyke trending E–W, (b) normal polarity direction obtained from baked host rock sample, (c) reversed polarity direction from dyke trending N–S, and (d) reversed polarity direction from baked host rock sample. Figures show stereographic projection, normalized intensity curve (J/J_0) and orthogonal projections of remanent magnetization directions.

300 °C to 585 °C (Figs. 6 and 11). It was mainly obtained from NE–SW trending dykes, which are more altered than other dykes (e.g. Pihlaja, 1987), and from coarse grained and/or weathered samples being more vulnerable to remagnetization. In some cases distinction between PEF and component S was not clear. The mean direction for component S is $D = 62.6°$, $I = 64.3°$ with $\alpha_{95} = 8.9°$ and it yields a paleomagnetic pole located at Plat = 53.3° N, Plong = 103.9° E ($A_{95} = 13.2°$) (Table 2).

4.2. Magnetic mineralogy

Observed Curie points and Verwey transition temperatures are listed in Table 3, and representative thermomagnetic (susceptibility versus temperature) curves are shown in Fig. 12. In most samples, high-temperature thermomagnetic curves showed two Curie temperatures. The higher Curie temperatures (T_{C2}) range from 574 °C to 584 °C in curves showing pronounced Hopkinson's peaks which have the typical feature of stable SD/PSD magnetite (Dunlop and Özdemir, 1997). Presence of magnetite is also supported by the well-characterized Verwey-transition at ~120 K during both susceptibility and remanent magnetization measurements. The lower-temperature Curie point and remanent magnetization measurements indicate the presence of maghemite, titanomagnetite or pyrrhotite. The majority of samples was saturated below 0.4 T during hysteresis measurements (Fig. 13; Table 3) supporting magnetite as the dominant magnetic mineral (O'Reilly,

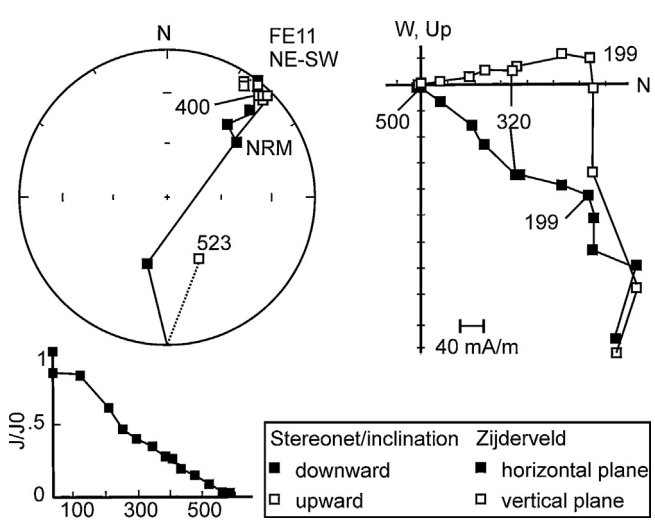


Fig. 5. Example of thermal demagnetization of NE–SW trending dyke showing primary component P. Figures show stereographic projection, normalized intensity curve (J/J_0) and orthogonal projections of remanent magnetization directions.

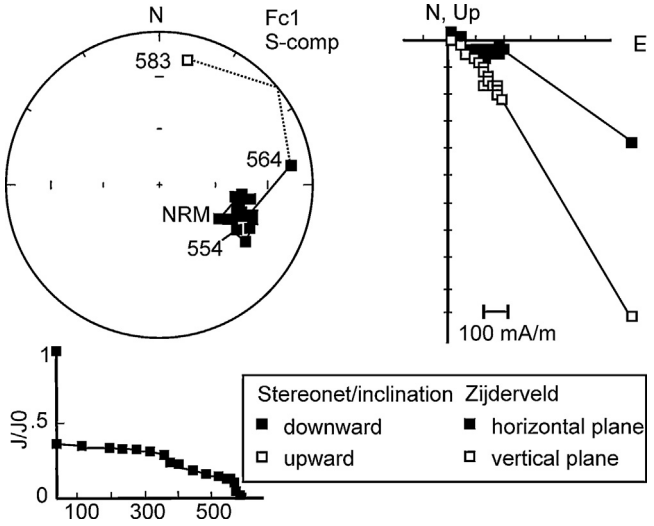


Fig. 6. Example of thermal demagnetization of NE–SW trending dyke showing secondary component S. Figures show stereographic projection, normalized intensity curve (J/J_0) and orthogonal projections of remanent magnetization directions.

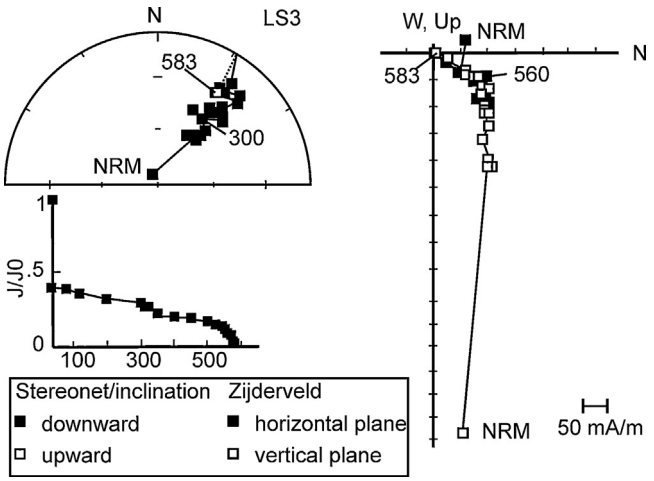


Fig. 7. Example of thermal demagnetization of the N–S trending dated dyke LS (1565 Ma; Lehtonen et al., 2003) showing a clear present earth field direction (PEF) component and also the primary component P. This sample is coarse grained and therefore carries strong PEF component.

1984; Dekkers, 1988). In the modified Day-plot (Day et al., 1977; Dunlop, 2002), most samples occur in the pseudo-single-domain (PSD) region (Fig. 13).

4.3. Anisotropy of magnetic susceptibility

The degree of anisotropy of magnetic susceptibility (AMS) ($P = K_{max}/K_{min}$) is low, and the shape parameter data are equally distributed showing both prolate and oblate signatures (Fig. 14) indicating that the characteristic remanent magnetization direction is not significantly affected by later deformation (Hrouda, 1982). In the studied dykes the magnetic foliation plane ($K_{max} - K_{int}$) agrees with the strike of the dykes (Fig. 15), indicating "normal" fabric (e.g. Knight and Walker, 1988; Rochette et al., 1992, 1999; Ernst and Baragar, 1992; Tauxe et al., 1998; Raposo et al., 2004; Raposo, 2011). This fabric has usually been interpreted to reflect the direction of magma flow, and the K_{max} inclination has been used to infer the relative position between magma source and dykes themselves (e.g. Knight and Walker, 1988; Raposo et al., 2004; Raposo, 2011). In some cases when unstable paleomagnetic directions or strong secondary components were observed (Fig. 15c), the magnetic foliation plane disagrees with the strikes of dykes (e.g. Fig. 15), most probably indicating later alteration processes.

4.4. Geochemistry

Fig. 16 shows that Nb concentrations of Satakunta dykes are within the range defined by other Subjotnian mafic dykes in Finland (e.g. Luttinen and Kosunen, 2006). Incompatible element ratios (Nb/Y) of Satakunta N–S and NE–SW trending dykes are similar to those of Subjotnian dykes in Åland (1.57–1.54 Ga), whereas Nb/Y ratios of E–W trending dykes show similarities to those of Häme dykes (1.67–1.64 Ga). These results support the grouping based on paleomagnetic data. The wide range of Nb/Y values (Table 2) indicates different geochemical features of the primary melts of these dikes (differences in the degree of melting and crustal contamination, and/or source compositions). Because Nb/Y ratios of the melt vary little during magma differentiation, the low Nb/Y values of the Satakunta N–S and NE–SW dykes and Åland dykes indicate they could originate from the same mantle source, but represent higher degree of melt compared with other Subjotnian mafic formations Table 4.

The grouping of Satakunta dykes based on their trend is also supported by Pb isotope compositions (Fig. 17, $^{207}Pb/^{204}Pb$ versus $^{206}Pb/^{204}Pb$ plot of Satakunta and Sipoo whole-rock mafic dyke samples). Both groups form isochrons, supporting the notion of closed systems. Combined Satakunta E–W and Sipoo Subjotnian

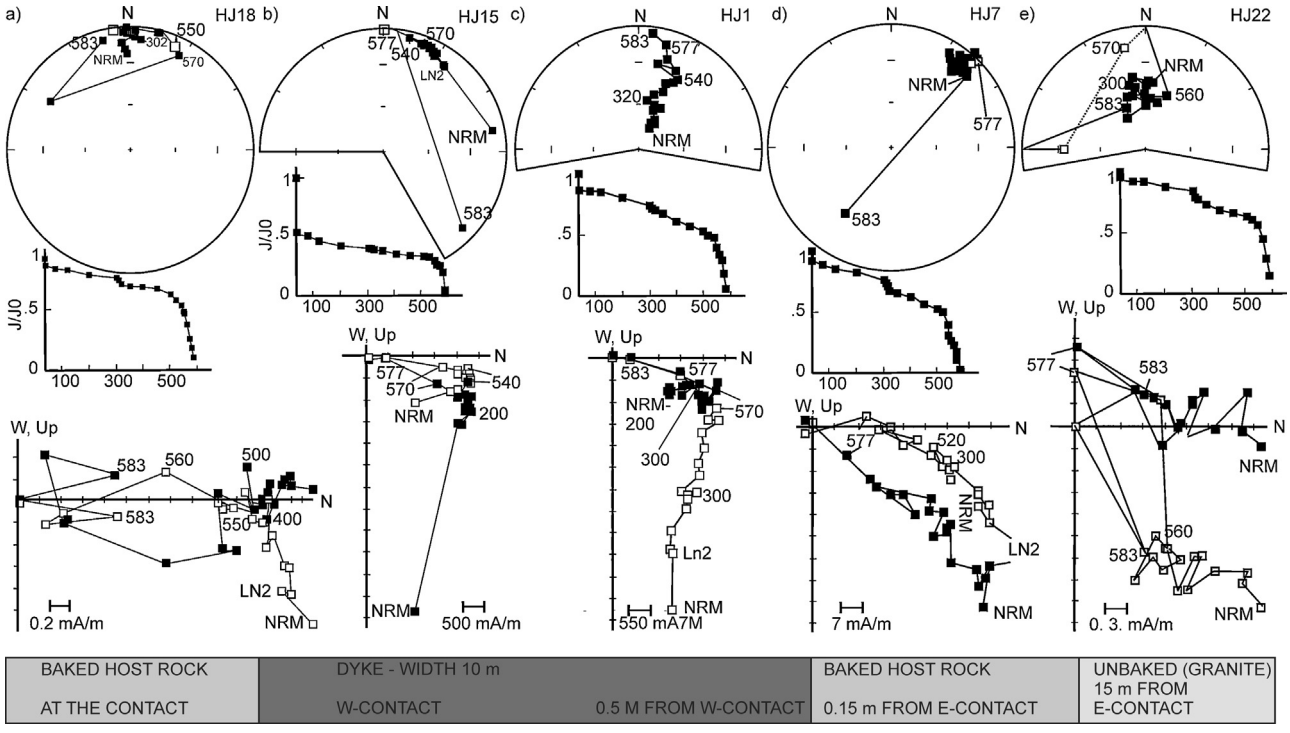


Fig. 8. Positive baked contact test for samples from the Satakunta N–S trending dyke (HJ), and its baked and unbaked granitic host rock. (a) Granite from right next to W-contact of the dyke, (b) dyke sample right at the E-contact, (c) dyke sample 0.5 m from contact, (d) baked host granite 0.15 m from W-contact, and (e) unbaked granite 15 m from contact.

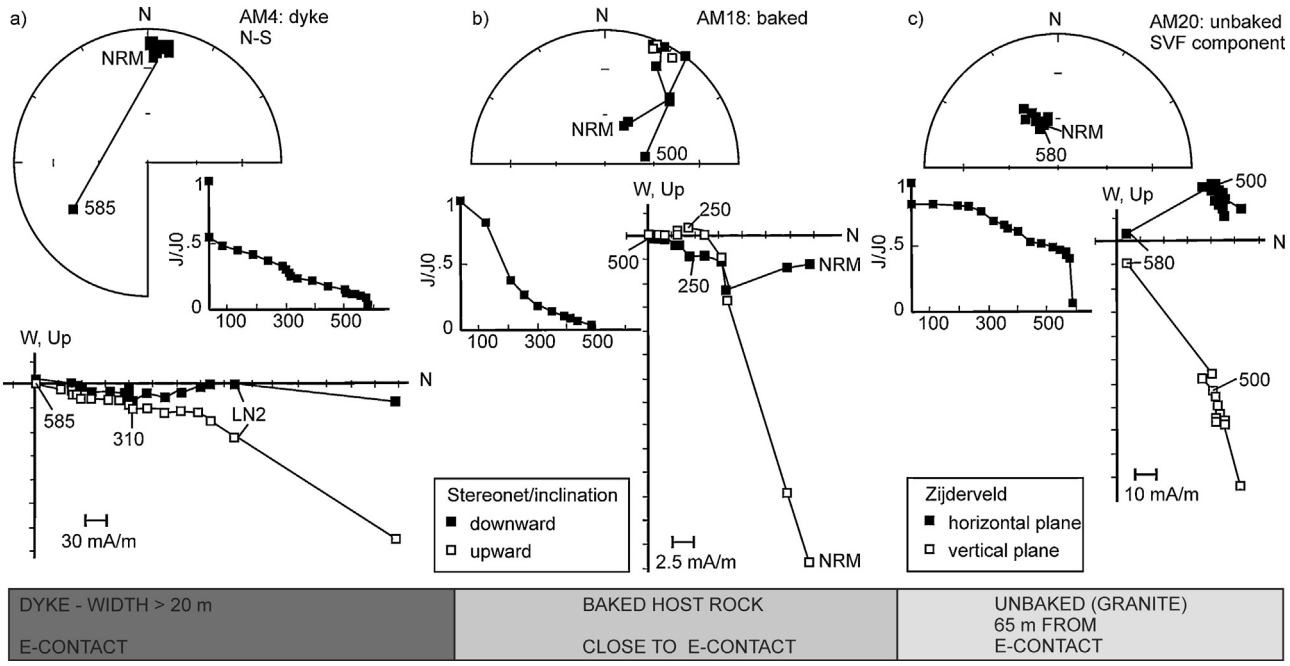


Fig. 9. Positive baked contact test for samples from the Satakunta N–S trending dyke (AM), and its baked and unbaked granitic host rock. (a) Dyke sample right at the E-contact, (b) baked host granite from E-contact, and (c) unbaked granite 65 m from contact.

dykes contain more radiogenic Pb (range in $^{206}Pb/^{204}Pb$ ratios from 16.795 to 20.927) than combined Satakunta N–S and NE–SW trending dykes (range in $^{206}Pb/^{204}Pb$ ratios from 16.608 to 18.002), suggesting different magma sources for these groups (Table 5).

5. Discussion

5.1. Quality of paleomagnetic poles from the Satakunta dykes

Half of the sampled Satakunta dykes show the shallow up- and/or downward pointing P-component, with both normal and reversed directions. The P-component was isolated in the E–W, N–S and NE–SW trending dykes with overlapping α_{95} error cones (Table 2; Fig. 10). The dyke swarms trending N–S and E–W show both normal and reversed directions, but they fail the positive reversal test (McFadden and McElhinney, 1990). Such non-antipodal directions are commonly observed in paleomagnetic records. Several possible hypotheses have been put forward to explain these records, including (1) the insufficient number of sites for the test; (2) the presence of a partially un-removed present-day field magnetization; and (3) changes in magnetic field geometry (e.g. the contribution of an axial octupole, Parés and Van der Voo, 2013). Rock-magnetic experiments show that P is carried by pseudo-single-domain magnetite with unblocking temperatures close to pure magnetite (Table 3; Figs. 14–15), which is not affected by later deformation (Fig. 16).

Geochemical evidence, field observations, and paleomagnetic data (Fig. 18) imply that the E–W trending dykes are older than the N–S and NE–SW trending dykes (Pihlaja, 1987). For example, in Fig. 16, the Nb/Y ratios and Nb concentrations in the N–S and NE–SW trending dykes lie in the same range as those in Åland dykes (1.54–1.57 Ga), and those in E–W trending dykes also sit in the same area as those in Häme and Suomenniemi dykes (1.64–1.67 Ga). Dyke swarms can also be separated based on Pb isotopes. It has

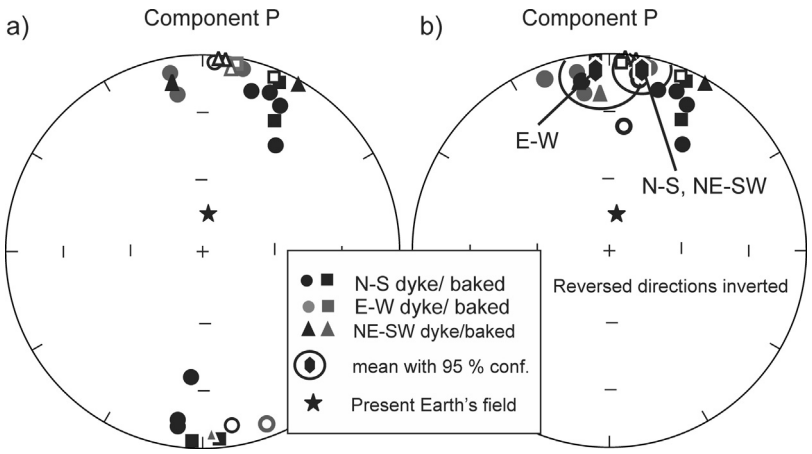


Fig. 10. Site mean directions for primary remanent magnetization for Satakunta dykes and their host rocks (see Table 2). (a) Site directions for component P and (b) site directions for component P with antipodal directions inverted. Hexagons represent the mean of both E–W and combined N–S and NE–SW trending dykes. Closed (open) symbols represent downward (upward) directions.

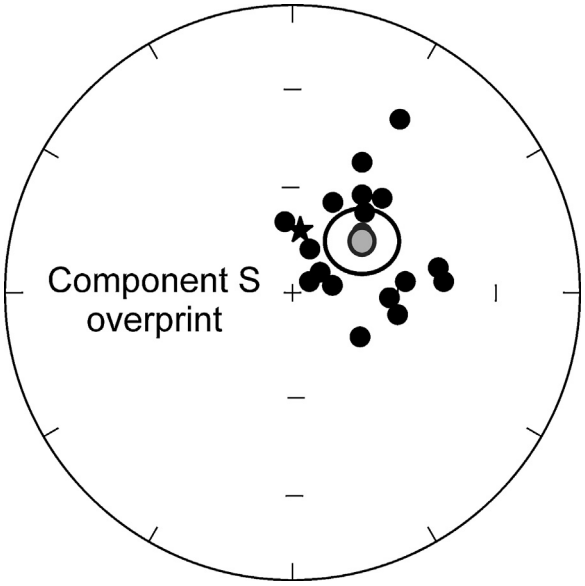


Fig. 11. Site mean paleomagnetic directions for interpreted secondary component S. Gray dot represents mean value. Closed (open) symbols represent downward (upward) directions. Black star represents Present Earth Field's direction on the sampling area.

been noted previously (L.J. Pesonen, unpublished data) that Subjotnian (ca. 1.57–1.54 Ga) mafic dykes in Åland Archipelago with reversed magnetization (Kumlinge-Brändö) are older than those with normal magnetization (Föglö-Sottunga). When comparing the position of normal (SK2(N)) and reversed (SK2(R)) poles (see Table 2) from E–W trending Satakunta dykes, pole SK2(R) plots closer to well-defined Svecofennian poles (1.88–1.79 Ga) than pole SK2(N), implying that SK2(R) could be older than the SK2(N). However, since none of these studied E–W trending dykes has been dated, we combine these poles and use that mean in further discussion. Furthermore, the positions of poles SK1(N) and SK1(R) do not indicate any clear age difference. Since only the normal polarity dyke (LS) has been dated, we will use the combined pole SK1 in further discussion. The new paleomagnetic poles are SK2 for E–W (SK2: Plat = 32.6° N, Plong = 205.5° E with A_{95} = 14.3°) and SK1 for combined N–S and NE–SW (SK1: Plat = 29.3° N, Plong = 188.1° E with A_{95} = 6.6°) trending dykes. When comparing the observed between-site dispersion (S) of site-mean paleopoles SK1 and SK2, we obtain "a" values between 11.4° and 15.7° (Table 2), which are consistent with those predicted (13–14°) from the secular variation Model G of Merrill and McElhinny (1983). This indicates that the geomagnetic secular variation has been adequately sampled and that the GAD was operating. The primary nature of both poles is demonstrated by positive baked-contact tests (Fig. 4). Since the error angles of poles SK1 and SK2 are overlapping we cannot truly distinguish their magnetization ages. However, pole SK2 is closer to older paleomagnetic key poles (with Svecofennian age, 1.9–1.8 Ga) than pole SK1 (Fig. 18), probably indicating that the pole SK2 is older than the pole SK1. Unfortunately there are no good quality paleomagnetic data coeval with the Häme dykes (1665–1635 Ma) for comparison. Paleomagnetism of Häme dykes has been reported by Neuvonen (1967), and one dyke belonging to the Häme swarm (Virmaila location) was studied by Salminen and Pesonen (unpublished data). Neuvonen (1967) reported a magnetization direction and pole similar to the secondary component of this study, whereas the Virmaila dyke shows a similar shallow normal polarity direction as the primary component found in this study. These results indicate that earlier methods were inadequate to isolate the primary component (e.g., Neuvonen, 1967 only found the secondary component). The younger Satakunta pole SK1 coincides with the pole of Föglö-Sottunga dykes (1540 ± 12 Ma and 1577 ± 12 Ma) and the error angles overlap with the pole from Bräven-Hälleforsnäs dykes (EW1; 1595 ± 3.5 Ma; 1602 ± 2 Ma, pole 17, Fig. 18), suggesting similar magnetization ages.

The pole SK1 fulfills six quality criteria of paleomagnetic poles (Van der Voo, 1990) providing a new key pole (e.g. Buchan and Halls, 1990; Buchan et al., 2000) for Baltica at 1565 Ma. The only criterion it does not satisfy is a similarity to a younger, but reversed, direction from Baltica in Carboniferous and Permian formations in Sweden and Norway (e.g. Torsvik et al., 1992). However, these areas are more than 500 km to the southwest of our sampling sites, and considering the positive baked contact tests for several dykes it is very unlikely that Carboniferous-Permian geological events generated the remanent magnetization P. Moreover, Pole SK2 satisfies five of Van der Voo's (1990) quality criteria, failing only the aforementioned similarity to younger poles from Baltica, as well as only a half-validated baked contact test, i.e., at one site the baked granite shows similar direction to the dyke, but the unbaked host rock was unstable. However, we have demonstrated that granites (~8 km from this site) carry primary magnetization (~1.88 Ga, Svecofennian).

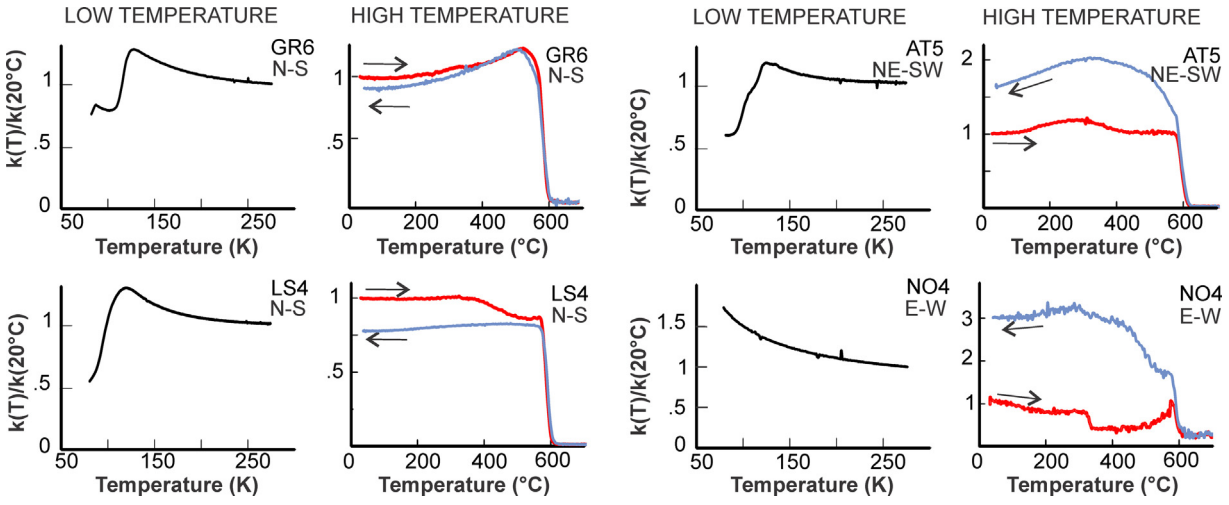


Fig. 12. Low and high temperature thermomagnetic curves showing variation in normalized magnetic susceptibility for the Satakunta dyke samples. Curves were corrected for furnace effects.

Table 3
Rock magnetic properties for dykes from Satakunta.

Susceptibility versus temperature						Hysteresis							Magnetization versus cold temperature						
Trend of dykes/sample	T_{v1} (K)	T_{v2} (K)	T_{C1} (°C)	T_{C2} (°C)	Notes	Mr (Am^2/kg)	Ms (Am^2/kg)	Hc (mT)	Hcr (mT)	Mrs/Mr	Hcr/Hc	S (T)	RT_{SIRM} cooling t_v (K)	RT_{SIRM} heating t_v (K)	$M_{300'}$ (%)	FC T_v (K)	ZFC T_v (K)	Magnetic minerals	Pmag component
N–S																			
AM1				574														mag	P_N
GR6	126.0			580	H	0.20	0.04	17.52	41.16	0.19	2.35	0.4						PSD mag	P_R
HH2				580	H	0.02	0.01	66.77	84.16	0.39	1.26	0.4						mag	P_N
IV7	121.0		282	582	H	0.64	0.09	9.90	3.47	0.14	0.35	0.2	116.7	124.9	68	119.4	118.5	PSD mag, mgh?	P_N and PEF/D
MA1	124.9			582	H, S	2.39	0.16	5.25		0.07		0.3						SD mag	P_N
LS4		116.3		580	H, c <h	2.33	0.12	4.71	14.31	0.05	3.04	0.4						PSD mag, mgh?	PEF/P_N
NE–SW																			
AT5	122.1	104.5	346	582														mag, mgh	P_N
FB1	122.7		342	579														mgh?, mag	PEF/S
FC2		118.8		575	H, S	0.08	0.01	10.35	30.18	0.13	2.92	0.6	115.9	128.5	44			PSD mag, mgh	*S*
FE3	121.6	105.1	304	584														PSD mag, mgh	P_N
FI4	123.2		326	584		1.04	0.06	4.90	14.01	0.06	2.86	0.2						PSD mag, mgh/Po?	*S*
FK7				582		0.18	0.02	12.62	29.84	0.13	2.37	0.3						mag	P_N
FS2	123.8		446	582														PSD mag, mgh	*S*
FT7	128.7		248	584	H	0.21	0.05	18.94	3.21	0.27	0.17	0.1						PSD mag, mgh	*S*
LK4	123.8		338	578	H, S	1.05	0.10	9.01	84.16	0.09	9.34	0.3	31.7/115.8	33.5/127.7	51	123.1	125.0	PSD mag, Po, mag	Unstable
E–W																			
AO3	127.5			580	H, S	0.91	0.13	12.50	22.77	0.14	1.82	0.4	117.1	127.1	52			PSD mag	P_N
BR5				540														mag	Unstable
FA2	123.8			580	S	1.42	0.11	6.01	13.51	0.08	2.25	0.7	124.4	127.1	48			PSD mag, Po(?)	P_N
NI1			320	580														mag, mgh	Unstable
NO4			332	576	H	0.02	0.01	23.47		0.45		0.2						PSD mag, mgh	P_N
NR2	124.9	108.9		579	H, S	2.12	0.21	9.71	12.64	0.10	1.30	0.5						PSD mag, mgh	*S*
OJ1				581		0.01	0.00	25.92	68.21	0.30	2.63	–						PSD mag	P_N
PR1	127.1			558														PSD mag	Unstable
SA1			300	537														Po/mgh?, mag	*S*

lat/long – sampling latitude and longitude, Tv1(2) – Verwey transition in Kelvins, $T_{C1/2}$ – Curie points in °C, notes: H – Hopkinson peak; S – shoulder type heating curve; $c < h$ – intensity during cooling is lower than during heating, Mr – saturation remanent magnetization is the maximum magnetization achieved by the specimen in the applied field, Ms – saturation magnetization is the magnetization remaining in the specimen after the field is switched off, Hc – coercive force, Hcr – coercivity of remanence, S – field needed to saturate sample, RTSIRM – 2.5 T SIRM given in room's temperature (300 K), FC – field-cooling in 2.5 T field, ZFC – zero-field-cooling, $M_{300'}$ – memory of the magnetization given in 300 K and then cooled down to 10 K and heated back to 300 K, SD – single-domain; PSD – pseudo-single-domain; MD – multi-domain, mag – magnetite, mgh – maghemite, Po – pyrrhotite, pmag component – obtained component during the paleomagnetic cleaning.

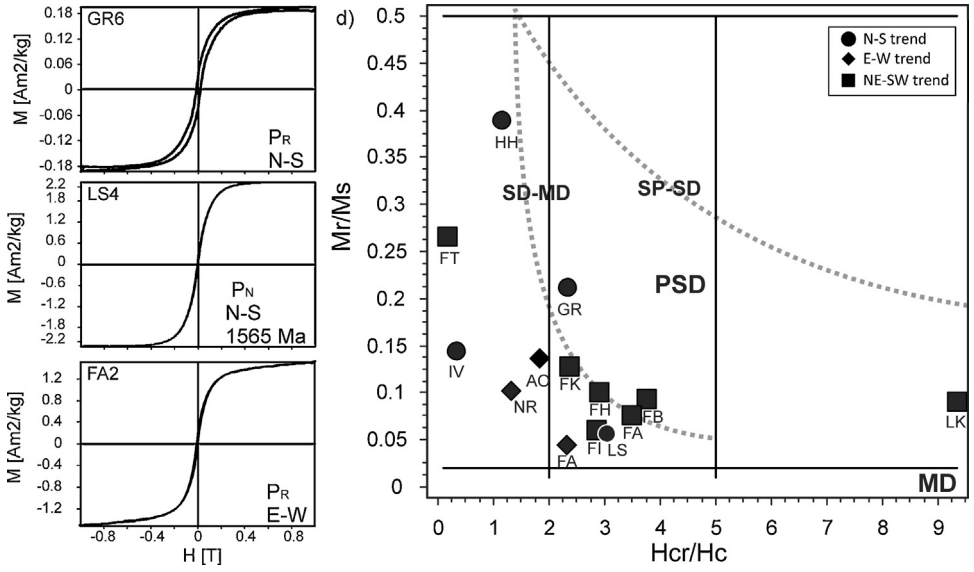


Fig. 13. (a)–(c) Examples of hysteresis loops for the selected Satakunta dykes. Trend of the dyke and obtained paleomagnetic component are indicated on lower right corner. (d) Hysteresis data for selected Satakunta dyke samples in Day plot (Day, 1977), which is modified after Dunlop (2002). Mr is saturation remanent magnetization; Ms is saturation magnetization; Hc is coercive force; and Hcr is coercivity of remanence.

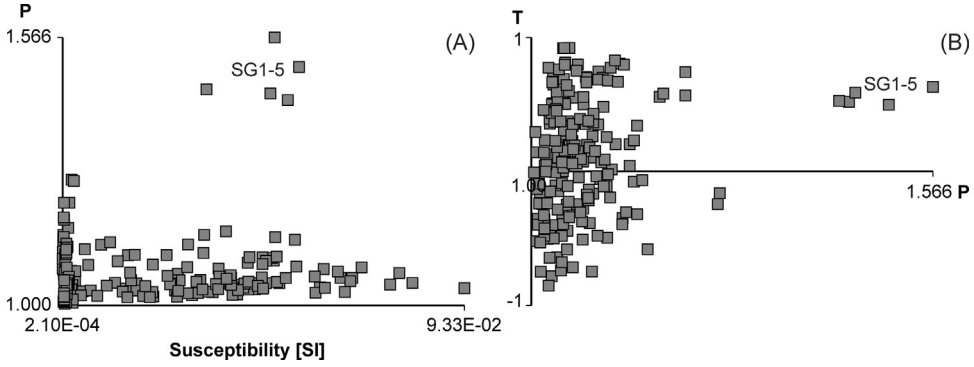


Fig. 14. (a) Degree of anisotropy (P) versus bulk susceptibility for all analyzed samples and (b) Jelinek's (1981) shape parameter (*T*) versus degree of anisotropy (*P*) for all analyzed samples.

5.2. Other Subjotnian paleomagnetic data for Baltica

When considering the quality of a paleomagnetic pole it is very important to know both the radiometric age of the rock unit and whether the obtained magnetization is primary. Thus, the numbers (1) and (4) of Van der Voo (1990) quality criteria are considered the most important ones. Many Subjotnian formations, mainly rapakivi granites and associated diabase and quartz-porphyry dykes, have been studied in Fennoscandia (e.g. Neuvonen and Grundström, 1969; Neuvonen, 1970, 1978, 1986; Bylund, 1985; Pesonen et al.,

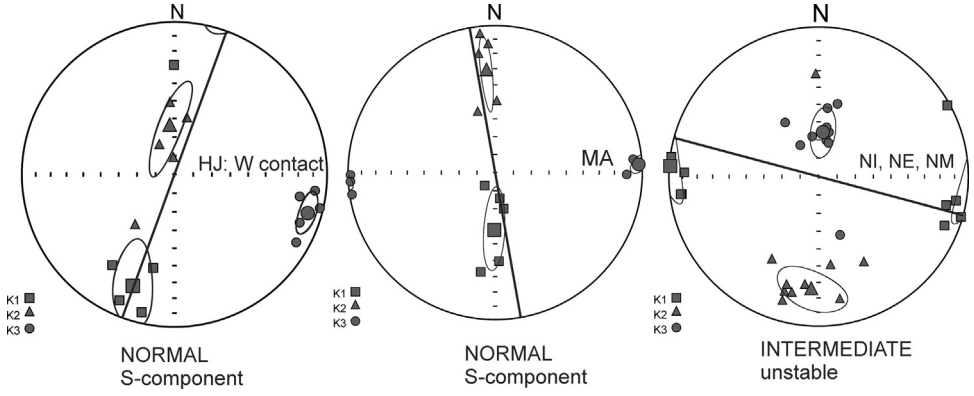


Fig. 15. Directions of susceptibility axes on a stereoplots. Examples of normal fabric, where magnetic fabric directions agree with dykes strike, are presented in (a) and (b) and intermediate fabric, where magnetic fabric direction disagree with dykes strike in (c). Symbols: K_1 (square) is maximum axis; K_2 (triangle) is intermediate axis; and K_3 (circle) is minimum axis. Black line represents the orientation (strike) of the dyke. The component obtained in paleomagnetic study is noted below stereoplots.

Table 4
Site mean anisotropy of magnetic susceptibility data for Satakunta diabase dykes.

Trend of dykes/Site	Lat(° N)/long(° E)	Strike	*N*	*k* (SI)	*L*	*F*	*P*	*T*	K_1 *D* (°)	K_1 I (°)	C_{1a}/C_{1b}	K_2 *D* (°)	K_2 *I* (°)	C_{2a}/C_{2b}	K_3 *D* (°)	K_3 *I* (°)	C_{3a}/C_{3b}	Fab.	Pmag
N–S																			
KA	62.301/21.379	325	5	0.047	1.035	1.026	1.061	−0.178	315	14	13/5	215	34	16/11	64	53	16/7	n	south
LS	62.206/21.429	360	5	0.069	1.028	1.037	1.066	0.129	348	14	15/4	232	61	16/2	85	25	7/4	n	PEF/P_N
VA	62.269/21.434	340	5	0.013	1.073	1.031	1.106	−0.396	118	17	14/4	23	16	15/2	251	66	6/4	n	South/S
HJ_E cont	62.059/21.670	20	5	0.038	1.017	1.021	1.038	0.148	358	14	28/2	88	4	28/5	193	75	7/3	im	P_N
HJ_W cont	62.059/21.670	20	5	0.045	1.007	1.038	1.046	0.686	201	23	28/9	354	65	28/6	106	10	9/6	n	P_N
IV	61.542/21.752	350	5	0.027	1.014	1.025	1.040	0.198	335	33	20/9	94	38	21/4	218	36	9/5	n	P_N/S
PL	62.279/21.431	345	5	0.042	1.024	1.037	1.062	0.191	203	56	55/7	346	28	55/10	85	17	13/9	n	Unstable
GR	62.240/21.380	360	5	0.040	1.063	1.064	1.130	0.031	325	37	12/6	193	42	26/7	77	27	26/11	n	P_R
HH	61.687/21.594	5	9	0.003	1.022	1.014	1.037	−0.192	66	44	32/15	234	45	41/18	330	6	35/14	n	Unstable
MA	61.726/21.609	350	5	0.048	1.019	1.076	1.096	0.596	181	58	24/6	355	32	24/3	87	3	7/3	n	P_N/S
SG	62.216/21.399	360	5	0.048	1.091	1.357	1.481	0.553	108	71	6/2	2	6	6/3	270	18	4/2	n	Unstable
OV: 3 dykes	62.154/21.395	10	8	0.036	1.025	1.025	1.050	0.107	305	83	37/18	167	5	35/22	77	5	35/32	n	P_R
NE–SW																			
FI;FS	61.680/21.542	40	6	0.026	1.031	1.037	1.069	0.085	229	17	15/9	101	63	13/8	326	20	16/8	n	*S*
FK	61.680/21.542	58	9	0.002	1.029	1.014	1.043	−0.336	359	2	57/25	268	24	57/18	93	66	26/17	n	*S*
FE	61.679/21.545	60	7	0.017	1.032	1.025	1.057	−0.127	229	27	15/5	108	46	16/10	338	32	14/9	n	P_N/*S*
FC	61.680/21.545	60	5	0.023	1.008	1.032	1.040	0.620	234	72	34/6	67	18	34/4	336	4	8/5	n	*S*
FF	61.687/21.594	40	5	0.009	1.096	1.050	1.153	−0.355	301	83	27/3	131	7	46/4	41	1	49/11	im	Unstable
FT	61.680/21.553	47	8	0.029	1.020	1.027	1.048	0.134	99	78	66/19	286	12	67/39	196	2	48/21	im	*S*
FD, FH	61.681/21.546	55	7	0.022	1.023	1.029	1.053	0.029	88	69	34/10	307	17	47/21	213	13	44/16	im	PEF/*S*
E–W																			
NI;NE;NM	61.670/19.843	280	10	0.001	1.032	1.038	1.071	0.069	274	1	17/7	184	24	18/11	6	66	14/7	im	Unstable
NO	61.560/21.820	330	7	0.001	1.037	1.035	1.073	−0.060	335	23	23/12	242	6	23/13	139	66	18/6	im	P_N
SA	61.744/21.616	270	6	0.001	1.028	1.054	1.083	0.103	286	7	25/20	178	67	57/21	19	22	57/12	n	*S*
NR	61.560/21.820	275	9	0.068	1.017	1.026	1.044	0.155	270	52	47/16	7	6	39/35	102	38	40/31	im	*S*
SL	62.197/21.441	270	5	0.001	1.014	1.103	1.118	0.750	89	52	15/8	268	38	15/7	358	1	11/2	n	SVF
RK	61.975/22.333	270	8	0.001	1.016	1.006	1.022	−0.408	1	64	7/6	142	21	48/6	238	15	48/6	im	*S*
FA	61.679/21.545	270	7	0.047	1.017	1.024	1.041	0.031	9	71	64/12	253	9	64/42	161	17	43/21	n	P_R/*S*
Mean			166	0.025	1.026	1.035	1.062												

Lat/long – sampling latitude and longitude, *N* – number of samples, *k* – bulk mean susceptibility, *L* – magnetic lineation: $L = K_1/K_2$, *F* – foliation: $F = K_2/K_3$, *P* – degree of anisotropy: $P = K_1/K_3$, *T* – Jelinek's shape parameter (Jelinek, 1981): $T = [2\ln(K_2/K_3)/\ln(K_1/K_3)] - 1$, K_1, K_2 and K_3 – mean AMS eigen vectors, which represent the maximum, intermediate and minimum susceptibility axis, respectively, $C_{1(2,3)a}/C_{1(2,3)b}$ – semi-angles of the major and minor axes of 95% confidence ellipse, Fab. – observed fabric: n, normal; im, intermediate, Pmag – obtained component during paleomagnetic cleaning.

Table 5
Whole rock Pb isotopic composition of Satakunta and Sipoo Subjotnian dyke samples.

Sample	Trend	Nb (ppm)	Y (ppm)	Nb/Y	$^{206}Pb/^{204}Pb$	Error 2σ	$^{207}Pb/^{204}Pb$	Error 2σ
Satakunta								
AM8	N–S	6.42541	16.16315	0.397534	17.43249	0.00179	15.51951	0.00169
AM15	N–S	14.53996	40.54537	0.35861	17.381173	0.00272	15.50314	0.00245
GR2	N–S	13.6871	34.78138	0.393518	16.842105	0.00175	15.44417	0.00158
GR6	N–S	14.74729	38.06752	0.387398	17.117234	0.00192	15.47661	0.00186
HJ3	N–S	11.55158	36.19871	0.319116	16.640045	0.00280	15.4201	0.00249
HJ8	N–S	12.13263	34.83948	0.348244	16.607516	0.00429	15.41815	0.00278
LS1	N–S	30.89705	86.18967	0.358477	16.772165	0.00152	15.44297	0.00160
OV2	N–S	11.02472	34.26518	0.321747	16.901321	0.00209	15.44494	0.00182
VA2	N–S	24.90825	53.07072	0.469341	16.658305	0.00125	15.43039	0.00115
AT3	NE–SW	34.60376	82.89722	0.41743	18.001753	0.00172	15.57205	0.00154
FB6	NE–SW	22.14079	45.46294	0.487007	16.876835	0.00182	15.44471	0.00178
FK4	NE–SW	18.51105	44.30201	0.417838	16.905494	0.00157	15.44861	0.00154
FK14	NE–SW	20.34388	41.10707	0.4949	16.724737	0.00178	15.42863	0.00178
AO3	E–W	5.845264	13.00106	0.449599	17.232005	0.00251	15.48936	0.00221
FA2	E–W	23.62465	47.88119	0.493401	16.795154	0.00177	15.43775	0.00171
FA8	E–W	22.16113	42.22913	0.524783	16.879112	0.00197	15.44857	0.00181
SL2	E–W	24.98991	29.33798	0.851794	18.357468	0.00318	15.61867	0.00277
Sipoo diabase								
SC1	E–W	44.64954	44.41855	1.0052	17.587254	0.00224	15.52636	0.00210
SC6	E–W	48.12103	57.9343	0.830614	17.708768	0.00210	15.54301	0.00184
SG4	NW–SE	47.6181	50.99451	0.933789	17.808986	0.00199	15.5479	0.00177
SG3	NW–SE	49.32342	47.21222	1.044717	17.473038	0.00182	15.51572	0.00166
SF3	NNE–SSW	53.31565	53.45127	0.997463	18.023488	0.00200	15.57932	0.00180
Sipoo quartz porphyry								
PN4	NW–SE	50.96362	123.0377	0.414212	20.438554	0.00127	15.83424	0.00117
PN8	NW–SE	53.18633	114.9975	0.4625	20.927274	0.00162	15.87693	0.00146
PV8	NW–SE	71.10264	137.0688	0.518737	19.455662	0.00155	15.66483	0.00135
PV9	NW–SE	70.42737	126.7407	0.555681	19.130062	0.00140	15.6659	0.00131
SA2	NW–SE	35.41246	63.67719	0.556125	17.272917	0.00121	15.51173	0.00119

1987, 1991; Pesonen, 1987; Bylund and Elming, 1992; Mertanen and Pesonen, 1995; Mertanen et al., 2008). These Subjotnian data are scattered, and A_{95} error angles of many of these poles are overlapping (Fig. 18b, Table 6). Some of them seem to be overprinted by later geological processes, and others do not have good age constraints. More importantly, some of them do not have positive baked contact test results. Only four Subjotnian poles for Baltica samples meet the two most important criteria (but may have problems in others): (1) Nordingrå granite, 1578 ± 19 Ma (Piper, 1980); (2) Kumlinge-Brändö dykes, 1571 ± 9 Ma (Pesonen and Neuvonen, 1981); (3) Satakunta SK1, 1565 Ma; and (4) Föglö-Sottunga dykes in Åland, 1540 ± 12 Ma and 1577 ± 12 Ma (Pesonen and Neuvonen, 1981). The pole from Nordingrå granite (1578 ± 19 Ma) is somewhat offset from the other ca. 1580 Ma poles (Fig. 18), which may arise from the fact that there is no control over the possible tilting of the granite at the time of remanence acquisition (Piper, 1980).

A closer look at Baltica's Subjotnian paleomagnetic data reveal that large scatter may be due to several reasons. As an explanation we favor unrecognized overprinting and/or unrecognized tilting of granites. An additional explanation, especially for the reversed polarity data, could be the simultaneous reversing of the Earth's magnetic field when the dykes acquired their magnetization. Scatter may arise also from individual plate motion (perhaps Baltica alone, or perhaps within the Nuna supercontinent; Zhang et al., 2012) or true polar wander at 1.6–1.5 Ga. Full understanding of these issues awaits further data collection and analysis.

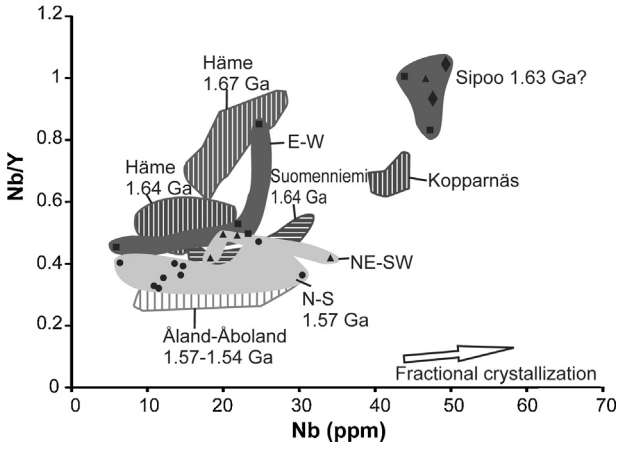


Fig. 16. Variation of Nb/Y versus Nb for the Satakunta and Sipoo (1633 Ma) Subjotnian dykes (Table 5). Compositional fields for other Subjotnian (Häme, Suomenniemi, Åland-Åboland, Kopparnäs) are shown for comparison (data from Luttinen and Kosunen, 2006). Fields of older (1.67 Ga) and younger (1.64 Ga) Häme dykes are shown separately.

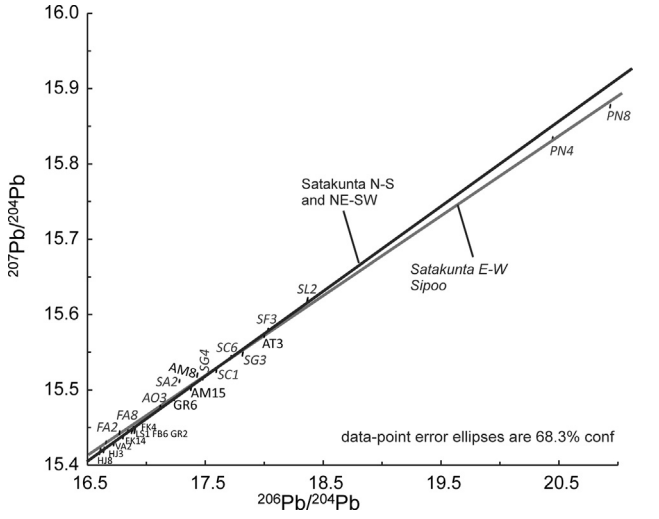


Fig. 17. Pb–Pb diagram showing the isotope ratios for measured Satakunta and Sipoo Subjotnian diabase dykes (Table 5).

Table 6
Selected paleomagnetic poles for Baltica and Laurentia.

Code	Pole/formation	Plat (° N)	Plong (° E)	A_{95}(°)	Euler pole (° N/° E/angle°)	RPlat (° N)	RPlong (° E)	Age (Ma)	1	2	3	4	5	6	Σ	Reference
Baltica-Fennoscandia																
1	Konchozero sill	−14	282	10				1974 ± 27	1	0	1	1	1	0	4	Pisarevsky and Sokolov (1999)
2	Mean Svecofennian 1	41	233	5				1881	1	1	1	0	0	0	3	Pesonen et al. (2003)
3	Mean Svecofennian 2	48	225	3				1840	1	0	1	0	0	1	3	Pesonen et al. (2003)
4	Shoksha sandstones	40	221	4	47/1.5/49.5[a]	13	266	1790–1770	0	1	1	1	1	1	5	Pisarevsky and Sokolov (2001)
5	Hoting gabbro[c]	43	233	11				1786 ± 10	1	1	1	1	0	0	4	Elming et al. (2009)
6	MEAN Ropruchey sill	39	217	7	47/1.5/49.5[a]	10	273	1770 ± 12	1	0	1	0	1	0	3	Pisarevsky and Sokolov (2001)
7	Sipoo diabase	32	184	31				1633?	1	0	1	0	1	0	3	Mertanen and Pesonen (1995)
8	Sipoo quartz porphyry	26	180	7	47/1.5/49.5[a]	11	233	1633 ± 10	1	1	1	0	1	0	4	Mertanen and Pesonen (1995)
9	SE Quartz porphyry dykes	30	175	9				1617 ± 2, 1639 ± 9, 1638 ± 53	1	0	1	0	1	0	3	Neuvonen (1986)
10	Hoting basic dykes[c]	21.9	146.7	13.7				1614 ± 24 (?)	1	0	1	1	1	0	4	Elming et al. (2009)
11	Nordingrå gabbro anorthosite	36.9	138.1	4.7				1578 ± 19	1	1	1	0	0	1	4	Piper (1980)
12	Nordingrå granite	32.4	149	10.5				1578 ± 19	1	0	1	1	0	1	5	Piper (1980)
13	Kumlinge-Brändö dykes[c]	12	182	7				1571 ± 9, 1576 ± 13	1	1	0	1	1	0	4	Pesonen and Neuvonen (1981)
14	Satakunta N–S and NE–SW: SK1[c]	29	188	7	47/1.5/49.5[a]	12	236	1565	1	1	1	1	1	1	6	This work
15	Satakunta E–W: SK2	33	206	14	47/1.5/49.5[a]	8	251	–	0	1	1	1	1	1	5	This work
16	Mean Satakunta combined	30	192	6	47/1.5/49.5[a]	11	240		1	1	1	1	1	1	6	This work
17	Bräven-Hälleforsnäs EW1	28	197	6				1595 ± 3.5, 1602 ± 2	1	0	1	0	1	0	3	Bylund (1985)
18	Bräven-Hälleforsnäs EW2[d]	62	176	11				1510–1560	1	0	1	0	1	0	3	Bylund (1985)
19	Föglö-Sottunga dykes[c]	28	188	9	47/1.5/49.5[a]	11	236	1540 ± 12, 1577 ± 12	1	1	0	1	1	0	4	Pesonen and Neuvonen (1981)
20	Ragunda dykes 2[d]	64.1	154.5	24.7				1505 ± 12, 1514 ± 5	1	0	1	0	1	1	4	Piper (1979)
21	Post-Ragunda dykes 1	14.6	194.4	11.1				1350–1520	0	1	1	0	1	1	4	Piper (1979)
22	Ragunda frm	51.6	166.6	7.1				1514 ± 5	1	1	1	0	0	1	4	Piper (1979)
23	Ragunda rapakivi	49.1	171	20.3				1503 ± 12, 1513 ± 9	1	0	1	0	0	0	2	Piper (1979)
24	Rödö dykes	41.6	201.7	10				–	0	1	1	1	1	0	4	Moakhar and Elming (2000)
25	Rödö granites[d]	6.2	175	5				1513	0	1	1	0	0	0	2	Moakhar and Elming (2000)
26	Salmi formation	6	200	11				1460	1	1	1	0	1	0	4	Shcherbakova et al. (2006)
27	Tuna dykes	21	171	10				1474 ± 4	1	1	1	0	1	0	4	Piper (1992)
28	Valaam monzodiorite	14	166	2				1458 ± 4	1	1	1	0	1	0	4	Salminen and Pesonen (2007)
29	Lake Ladoga mafic rocks[c]	15	177	6	47/1.5/49.5[a]	5	220	1452 ± 12	1	1	1	1	1	1	6	Lubnina et al. (2010)
30	Mean post-Jotnian intrusions[c]	4	158	4	47/1.5/49.5[a]	6	198	1265	1	1	1	1	1	0	5	Pesonen et al. (2003)
Laurentia																
Cle	Cleaver dykes[c]	19	277	6				1740 +5/−4	1	1	1	1	1	0	5	Irving et al. (2004)
Dub	Dubawnt group[c]	7	277	8				1785 ± 4	1	1	1	1	1	1	6	Park et al. (1973)
Mel	Melville Bugt dyke swarm	5	274	9	67.5/241.5/−13.8[b]	2	262	1622 ± 3, 1635 ± 3	1	1	1	0	1	1	5	Halls et al. (2011)
WCD	Western Channel diabase[c]	9	245	7				1592 ± 3, 1590 ± 4	1	1	1	1	1	0	5	Irving et al. (1972), Hamilton and Buchan (2010)
	Michikamau intrusion[c]	−2	218	5				1460 ± 5	1	1	1	1	0	1	5	Emslie et al. (1976)
	St. Francois Mnt acidic rocks[c]	−13	219	6				1476 ± 16	1	1	1	1	1	0	5	Meert and Stuckey (2002)
Mean	Mean	−7	218	6				1468								Interpolated
Mac	MacKenzie dykes grand mean[c]	4	190	5				1267 + 7/−3	1	1	1	1	1	0	5	Buchan and Halls (1990)

Code – code in Fig. 19; Plat – pole latitude; Plong – pole longitude; A_{95} – Fisher statistics parameter. RPlat(RPlong) – rotated pole latitude(longitude).

[a] Used rotation of Evans and Pisarevsky (2008) for Baltica to Laurentian reference frame.

[b] Used rotation of Roest and Sirvastava (1989) for Greenland to Laurentian reference frame.

[c] Primary magnetization demonstrated by baked contact test from the dated rock units.

[d] Suspected overprinted magnetic direction.

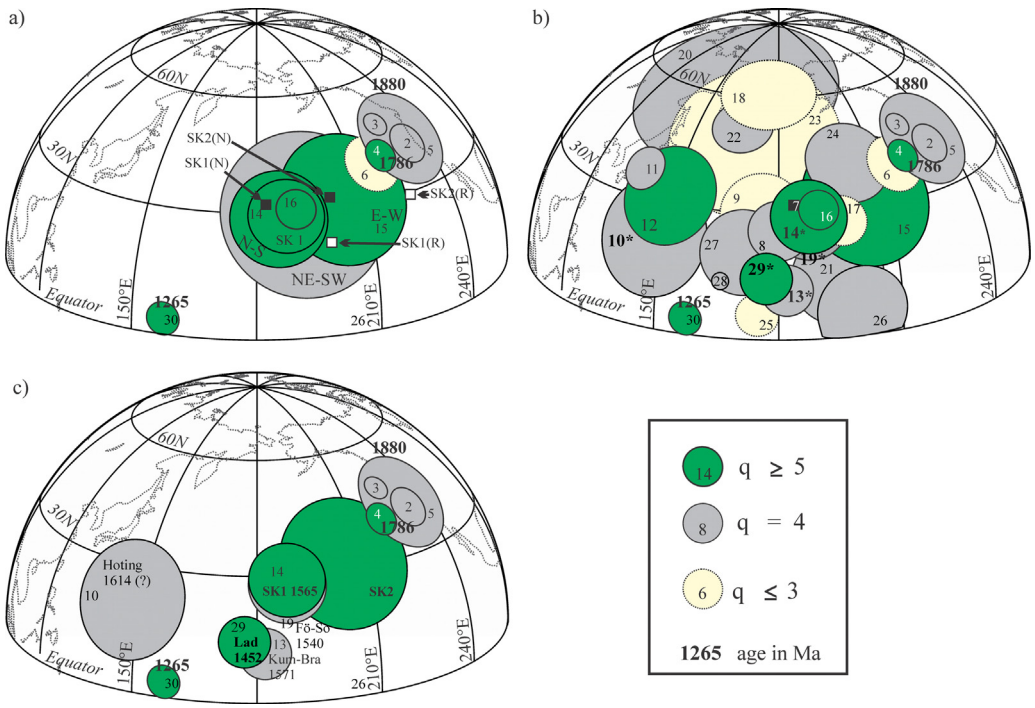


Fig. 18. (a) Satakunta paleomagnetic poles from dykes by different trends with relevant Precambrian poles for Baltica. Normal (black, N) and reversed (white, R) data for SK1 and SK2 poles are presented as squares without error bars. See coding in Tables 2 and 6, (b) poles from this study with other Subjotnian and relevant Precambrian paleomagnetic poles of Baltica. We have marked with * the most reliable Subjotnian poles (see Section 5.2), and (c) selected poles of Baltica with $q \geq 4$. See numbering in Table 6.

5.3. *Long-lasting NENA configuration within supercontinent NUNA*

Based on a tight cratonic fit, concordance with basement geology, correlations of geochronology, and matching of recent paleomagnetic poles, some authors favor the long-lasting (1.8–1.2 Ga) configuration of Baltica and Laurentia in the NENA (North Europe-North America) configuration whereby eastern Greenland is juxtaposed with the Timanide-Uralide margin of Baltica (e.g. Gower et al., 1990; Buchan et al., 2000; Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008; Pisarevsky and Bylund, 2010) forming the core of the supercontinent Nuna (see

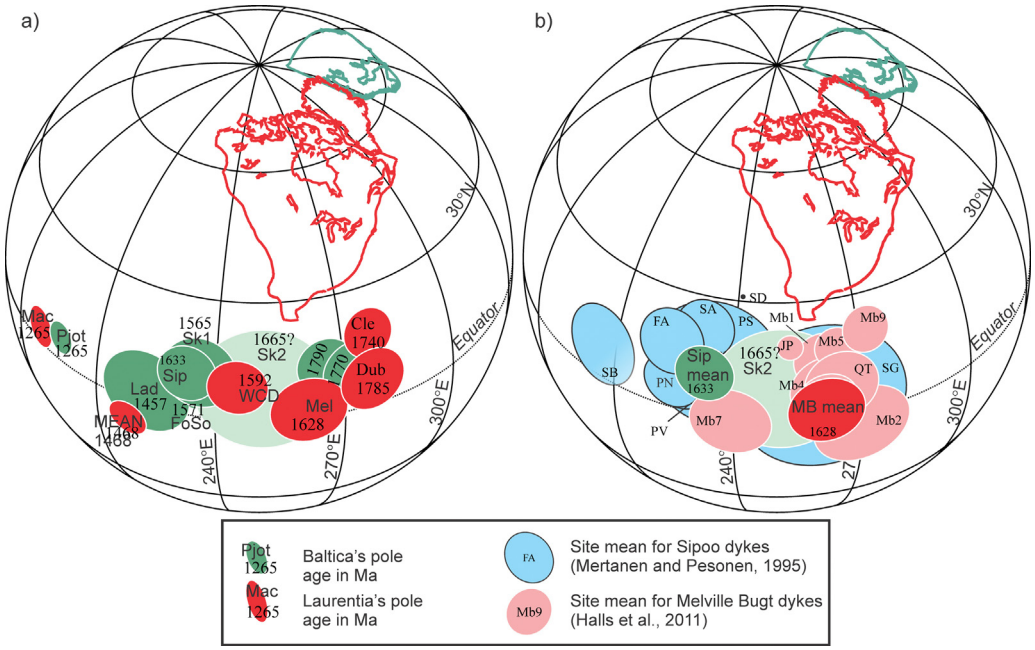


Fig. 19. Testing the NENA fit. (a) High quality poles of Laurentia and Baltica rotated to NENA configuration. Used poles are listed in Table 6, (b) virtual geomagnetic poles for Melville Bugt dykes, Greenland (normal polarity is used; Halls et al., 2011) and for Sipoo dykes, Finland (Mertanen and Pesonen, 1995). Additionally SK2 pole of this study is plotted. Baltica and its poles are rotated to reference frame of Laurentia by Euler pole: Plat = 47°, Plong = 1.5°, angle = 49° (Evans and Pisarevsky, 2008). Poles from Greenland are rotated to reference frame of Laurentia by Euler pole: Plat = 67.5°, Plong = 241.5°, angle = −13.8° (Roest and Sirvastava, 1989).

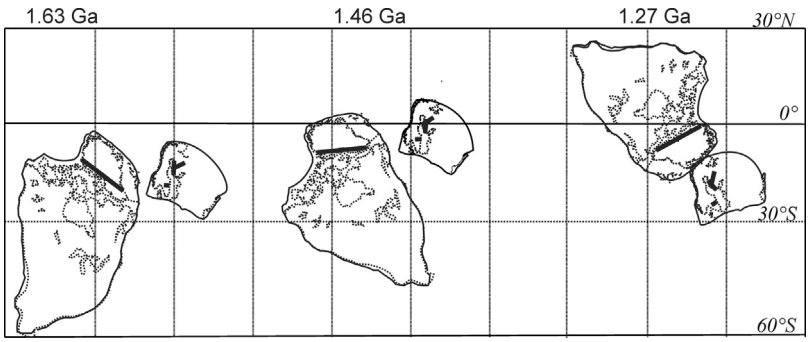


Fig. 20. Reconstructions of Baltica (B) and Laurentia (L) at 1.63 Ga, and 1.46 Ga suggested by Halls et al. (2011) that differ from proposed long lasting NENA fit (Gower et al., 1990; Patchett et al., 1978; Piper, 1980; Pesonen et al., 2003; Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008; Pisarevsky and Bylund, 2010; Pesonen et al., 2012) and reconstruction at 1.27 Ga (Pesonen et al., 2003). For 1.63 Ga – B: reinterpreted data from Sipoo porphyry dykes, Finland (Halls et al., 2011). L: location of Laurentia based on an adjustment of the Melville Bugt (Greenland) pole position that lies within its 95% confidence circle (Halls et al., 2011). For this reconstruction Melville Bugt pole from Greenland was at first rotated to reference frame of Laurentia (see Table 6). In this reconstruction the used polarity for Melville Bugt and Sipoo are different. For 1.46 Ga – B: combined Lake Ladoga area and Valaam sill (1452 ± 12 Ma) pole, Russia (Salminen and Pesonen, 2007; Lubnina et al., 2010). L: mean pole 1460 Ma (calculated in Salminen and Pesonen, 2007). For 1.265 Ga – B: mean pole from Postjotnian mafic dykes (1.265 Ga), Finland (Pesonen et al., 2003). L – MacKenzie dykes (1.265 Ga), Canada (Buchan and Halls, 1990) in NENA fit. Black line in Greenland (Baltica) cartoon indicates Melville Bugt dykes (Häme, Satakunta, and Bräven-Hälleforsnäs dykes related to rapakivi magmatism).

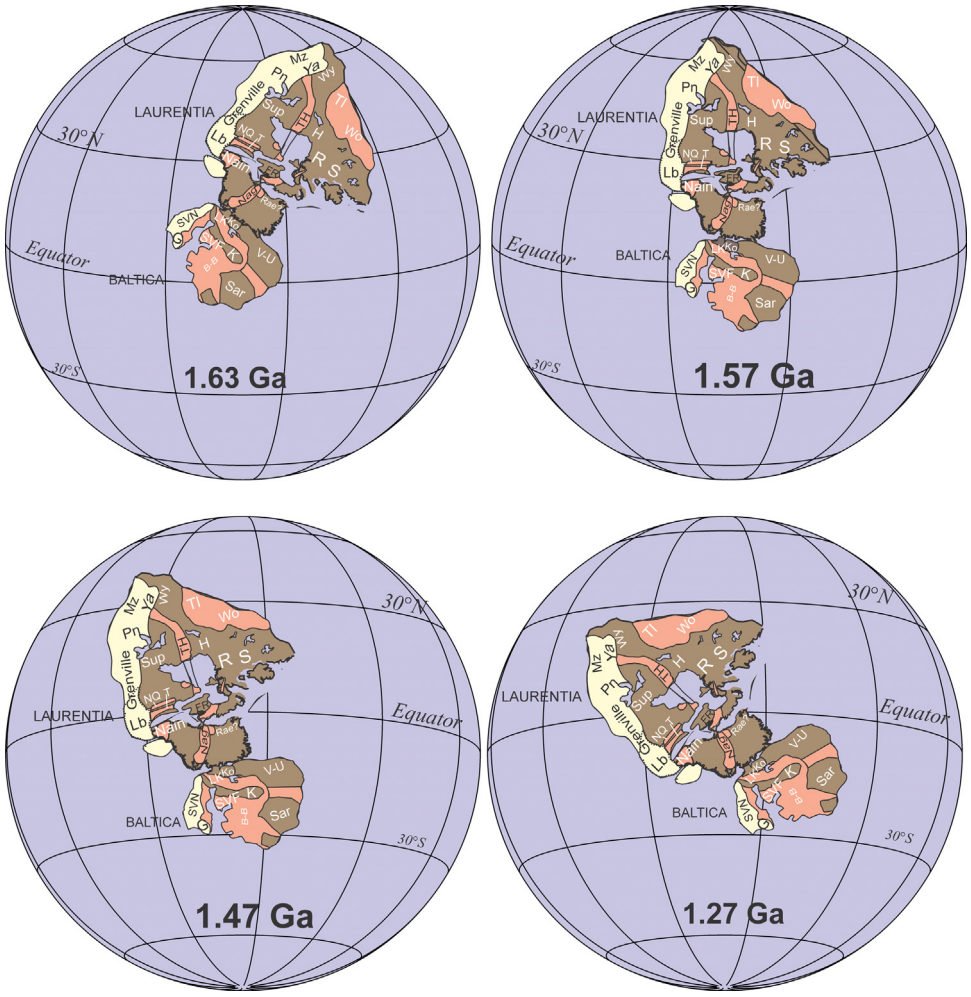


Fig. 21. Reconstruction of Baltica and Lauretina between 1.63 Ga and 1.27 Ga forming the core of NENA. Data are listed in Table 6. Archean cratonic areas: Baltica (Ko – Kola; K – Karelia; V-U – Volgo-Uralia; Sar – Sarmatia); Laurentia: (S – Slave, Sup – Superior, H – Hearne; R – Rae; Wy – Wyoming; Nain in Greenland). Proterozoic: Baltica (SVF – Svecofennian domain; LK – Lapland-Kola; G – Gothian; B-B – Baltic-Belarus; SVN – Sveconorwegian); and Laurentia (Pn – Panokean; Mz – Mazatzal; Ya – Yavapai; TH – Trans-Hudson; Tl – Taltson; Wo – Wopmay; NQ – New Quebeck; T – Torngat, FR Foxe-Rinklan; Lb – Labradorian; Nag – in Greenland).

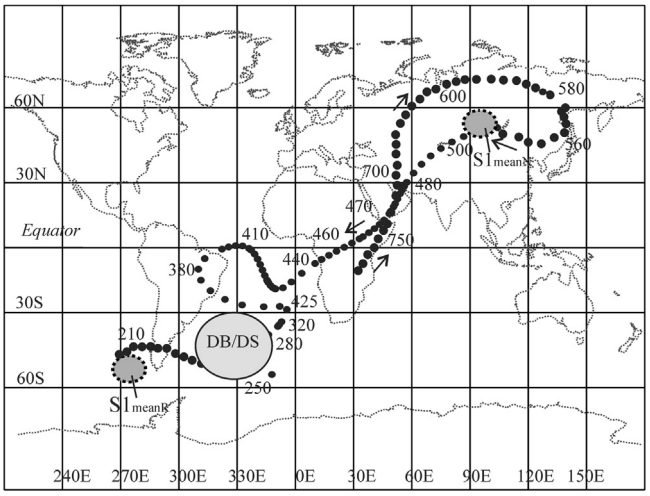


Fig. 22. Secondary pole from Satakunta dykes plotted on the Paleozoic apparent polar wander path of the Fennoscandian shield (Torsvik et al., 1996). Secondary DB/DS pole from Olkiluoto (Mertanen et al., 2008). Numbers indicate years in Ma.

also Hoffman, 1989, 1996; Åhäll and Connelly, 1998; Meert, 2002; Rogers and Santosh, 2002; Zhao et al., 2004; Condie, 2004; Pesonen et al., 2012). Paleomagnetic poles can be used to test the duration of a particular continental configuration by rotating coeval poles to the appropriate configuration (e.g. Evans and Pisarevsky, 2008). If the poles overlap there is paleomagnetic evidence consistent with that configuration for the proposed time interval. We have rotated selected key paleomagnetic poles of Baltica to the present reference frame of North America in the NENA configuration, using Euler parameters of Elat = 47.5°, Elong = 1.5°, angle = 49° (Evans and Pisarevsky, 2008; Table 6). In this reconstruction the combined mean pole of Satakunta (1565 Ma) overlaps with the 1592 ± 3 Ma Western Channel Diabase (WCD) pole (Fig. 19a) indicating that the NENA configuration is permissible based on paleomagnetic data. NENA is also valid at 1.75 Ga, at 1.46 Ga and at 1.27 Ga (Fig. 19). In the NENA fit, Archean to Paleoproterozoic formations are continuous from North America through Greenland to northern Baltica. For example, these fits allow correlations of subduction-related magmatism at around 1.7 Ga, stretching from the 1.8–1.7 Ga Yavapai and 1.7–1.6 Ga Mazatzal provinces in southwestern Laurentia, through the 1.7–1.6 Ga Labradorian province of eastern Laurentia, to the 1.7–1.5 Ga Gothian province of southwestern Baltica. Likewise, the trends of coeval 1.27 Ga MacKenzie dykes in Laurentia and Post-Jotnian dykes in Baltica become contiguous in these reconstructions (see also Pesonen et al., 2003; Elming and Mattsson, 2001). The well-known 1.27 Ga reconstruction avoids polarity ambiguity since the data from mafic dykes of both Laurentia and Baltica are of single polarity (Pesonen et al., 2003).

Despite the geological and paleomagnetic evidence for the joint history between 1.8 Ga and 1.27 Ga, recent mean paleomagnetic data from Greenland appear to argue against the NENA configuration. In detail, the paleomagnetic pole of Melville Bugt dykes (Greenland; 1622 ± 3 Ma, 1635 ± 3 Ma; Halls et al., 2011) does not overlap with the coeval pole of Sipoo dykes (Finland; 1633 Ma; Mertanen and Pesonen, 1995) in the NENA reconstruction (Fig. 19a), but it overlaps with the Satakunta SK2 pole (E–W dykes). Moreover, if the Satakunta SK2 pole from this study represents that having the same age as the Häme dykes (1665–1635 Ma; Vaasjoki et al., 1991), then it supports the NENA fit at 1.63 Ga (Fig. 19). However, the individual site-mean data from both Melville Bugt and Sipoo dykes show a large scatter between the virtual geomagnetic poles of both continents, which does not exclude the possibility that continents were together at 1.63 Ga (Fig. 19b).

In producing paleomagnetic and geochronological data from Melville Bugt dykes, Halls et al. (2011) proposed a continental configuration whereby Laurentia and Baltica are juxtaposed so that the southern end of the Melville Bugt dyke swarm can be projected toward the Fennoscandian rapakivi province (Fig. 20). Based on the proximity and geochemical data, Halls et al. (2011) suggest that Melville Bugt dykes in Greenland are fed by the source below the continental crust underneath Fennoscandian rapakivi province, which is also the source for the Fennoscandian rapakivi-related magmas. However, there are a few concerns with their reconstruction. First, the use of different polarity options for coeval paleopoles from Laurentia and for a modified pole of Baltica (Sipoo dykes, Finland – 1633 Ma) could be problematic of matching the well-known 1.26 Ga reconstruction that has no polarity ambiguity (Pesonen et al., 2003). Second, the Subjotnian dykes in Fennoscandia do not support the same magma source for Melville Bugt and Fennoscandian dykes for four reasons: (1) the variations in major elements and REE geochemical compositions of Subjotnian dykes in Finland (such as Häme and Åland archipelago dykes) are large and outside the geochemical range of Melville Bugt dykes (Luttinen and Kosunen, 2006); (2) recent geochemical data indicate different sources for Subjotnian dykes within Finland (Salminen, unpublished data); (3) the widths of the dykes in Fennoscandia are so narrow that they could not have been able to feed the huge Greenland dykes; and (4) the age distribution of dykes in Fennoscandia do not support the same magma source with Greenland dykes. The age of the Sipoo dykes (Southeast Finland) agrees with Melville Bugt dykes, but other Subjotnian dykes in between these intrusions, in southwest Finland and in Sweden, are younger. Consequently, we propose that the NENA fit, where Baltica occupied low latitudes, shown in Fig. 21, was valid also at 1.63–1.56 Ga, and that it formed the core of Nuna during 1.8–1.2 Ga. The Mesoproterozoic 1.5–1.3 Ga red-bed sedimentation (for example the Satakunta sandstone) also supports this low-latitude location of Baltica in Subjotnian times (Rämö et al., 2005; Klein et al., in this volume).

5.4. Secondary magnetization, Pole S

Fifteen sites show a secondary component S as a lower unblocking temperature overprint (Table 2; Fig. 11). This component S has pole: Plat = 53.3° N, Plong = 103.9° E, A_{95} = 13.2° (Table 2) and is obtained mainly from the weathered NE–SW trending dykes and in the samples taken from the coarser grained interior of the dykes. It has been suggested that NE–SW and NW–SE trending brittle fractures, which formed during the NW–SE directed stress field that caused normal faulting on Svecofennian domain, have been most vulnerable to later reactivation and remagnetization (Elminen et al., 2008; Mertanen et al., 2008). In the case of Satakunta dykes, this is supported by petrophysical and rock magnetic analyses, which show that remanent magnetization values are enhanced and that maghemite is present more often in samples showing component S than in samples showing primary component P, thus favoring hydrothermal origin for the secondary component S.

The secondary remanent magnetization direction similar to component S, with northeasterly declinations, has been widely observed all over the Fennoscandian shield (e.g. Bylund, 1985; Bylund and Elming, 1992; Mertanen and Pesonen, 1995; Mertanen, 2008; Preeden et al., 2009). The direction of this component in Subjotnian formations is always close to PEF direction and usually comes from the most altered dykes. We have plotted secondary poles from Satakunta dykes in Fig. 22 on the Paleozoic apparent polar wander path of Baltica. Normal polarity poles are on the Cambrian part of the APWP, and reversed polarity poles on the early Mesozoic part of the APWP. In the case of the early Mesozoic ages, this component could be related to early stages in the

breakup of Pangea supercontinent. If the component had Cambrian age, then it could reflect distal expressions of the Finnmarkian orogeny (Roberts, 2003). Since this component is mainly carried by maghemite, we favor the early Mesozoic age and suggest that it is caused by hydrothermal fluids activated by the break-up of Pangea (see Preeden et al., 2009).

6. Conclusions

The studied Subjotnian diabase dykes, associated with rapakivi magmatism in the Satakunta area in Finland, were divided into two groups based on their trends, and paleomagnetic and geochemical results. After removal of a secondary component of possible early Mesozoic age, a two-polarity stable component is revealed in each of the two groups of dykes. Rock-magnetic analyses and positive baked-contact tests indicate that both of these components represent primary thermoremanent magnetizations. The new key pole SK1 (Plat = 29.3°, Plong = 188.1°, A_{95} = 6.6°) was derived from N–S (1565 Ma, Lehtonen et al., 2003) and NE–SW trending dykes, and it fulfills six quality criteria for paleomagnetic poles (Van der Voo, 1990). E–W trending dykes provided primary paleomagnetic pole SK2 (Plat = 32.6°, Plong = 205.5°, A_{95} = 14.3°) that meets five quality criteria. Since the error angles of these poles overlap, paleomagnetism cannot truly distinguish the magnetization ages of these poles. However, the relative position of the poles compared to other well-defined paleomagnetic poles of Baltica indicates that SK2 is older than SK1. Comparison of the new Satakunta poles with the present Mesoproterozoic paleomagnetic data and correlations of geochronology and basement geology for Baltica and Laurentia, suggest that the NENA configuration is valid during the whole time period between 1.8 and 1.2 Ga, forming part of the center of Nuna supercontinent.

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